NODE ANALYSIS METHOD

CIRCUIT ANALYSIS METHOD

prepared by:

Gyro A. Madrona

Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

LABORATORY

50%

EXAM

PENALTY

0

ABSENT

3

45-min LATE

5 max

15-min LATE

TOPIC OUTLINE

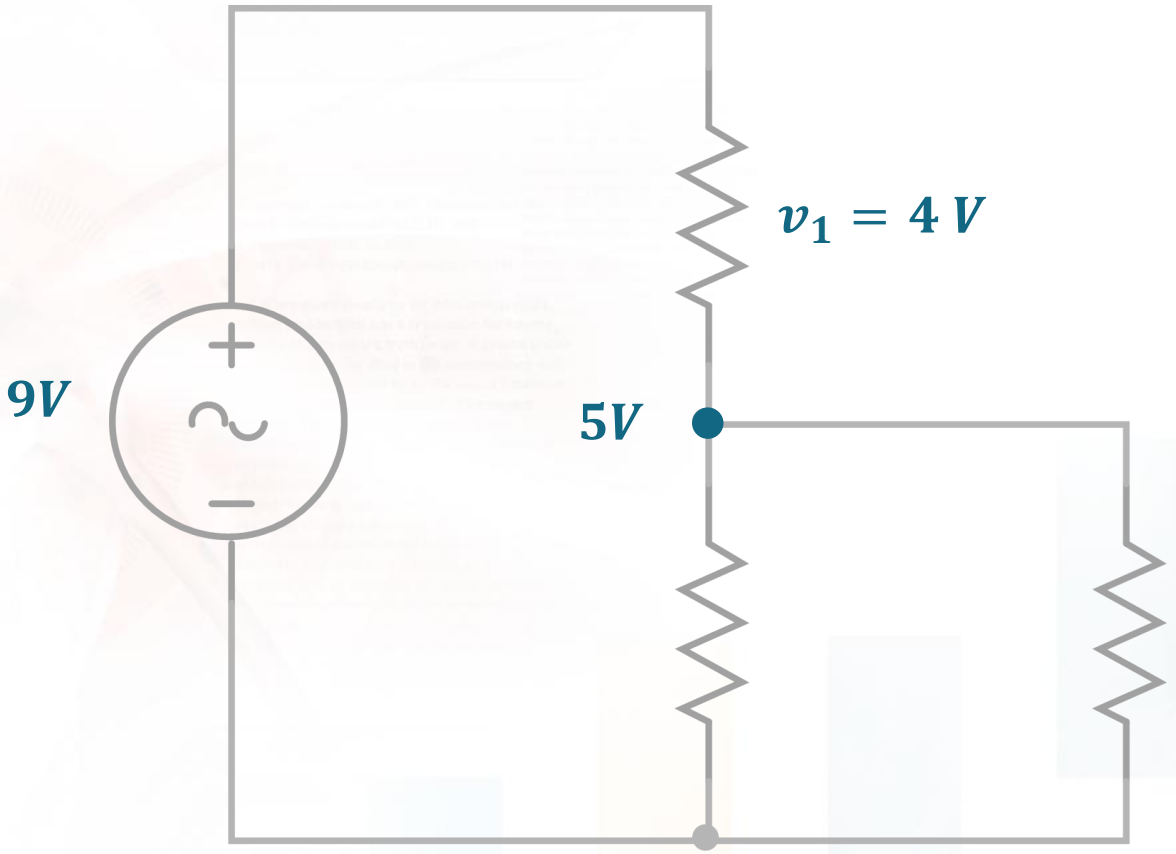
Node Voltage

Node Analysis Method

NODE VOLTAGE

NODE VOLTAGE

The voltage **difference** between two points, or **nodes**, in an electrical circuit



NODE ANALYSIS METHOD

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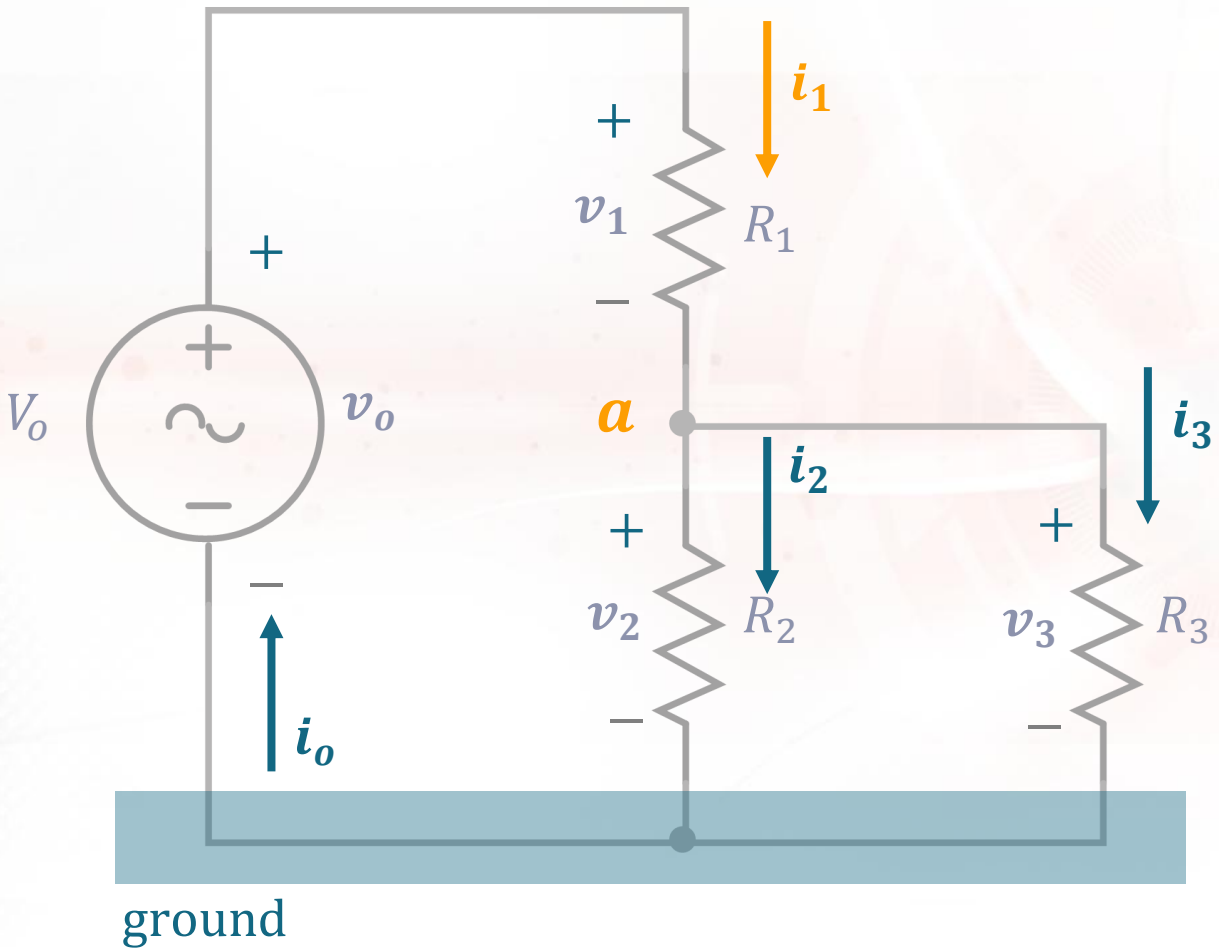
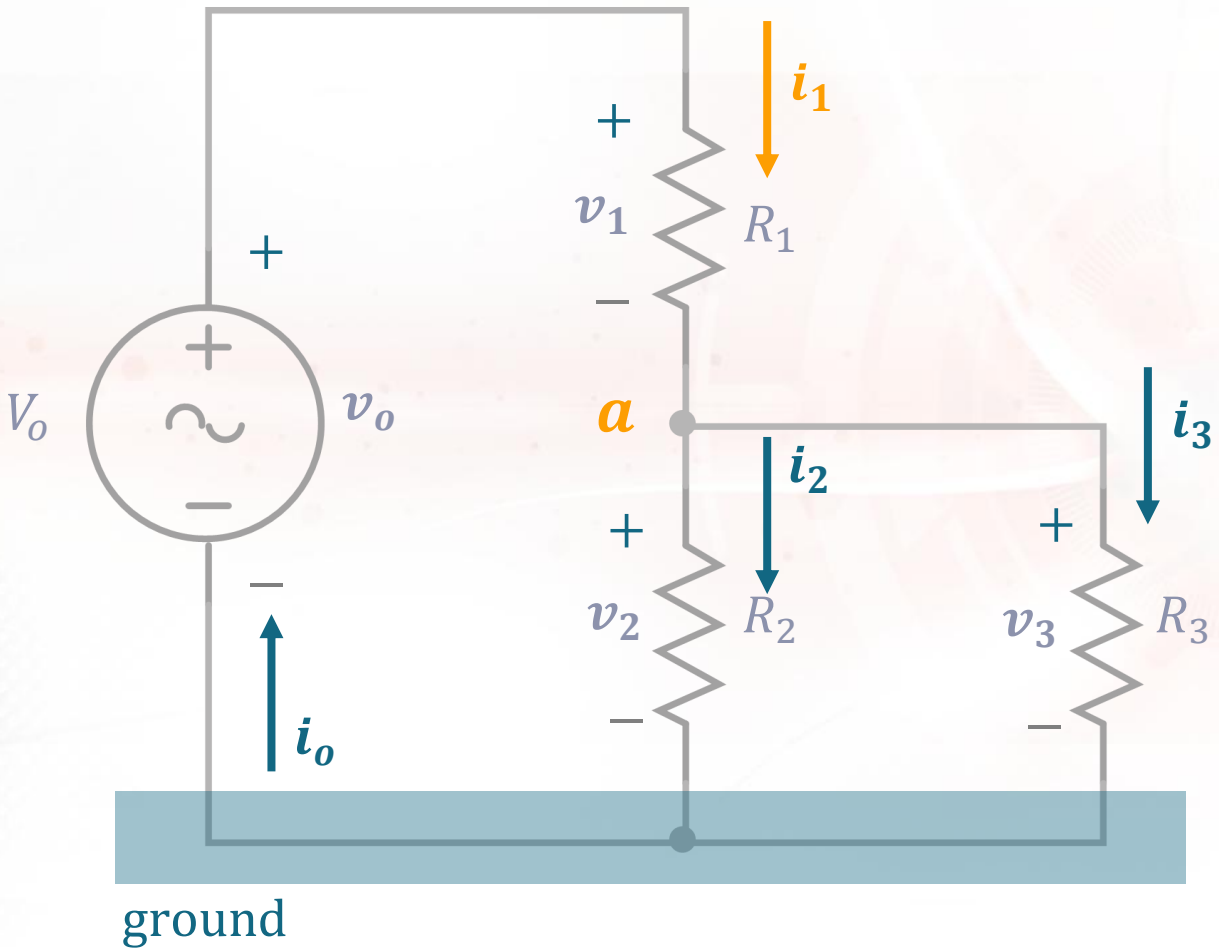


Uses **KCL** implicitly to formula $\boldsymbol{vi}$ relationship equation

KCL @ a

$$-i_1 + i_2 + i_3 = 0$$

NODE ANALYSIS METHOD

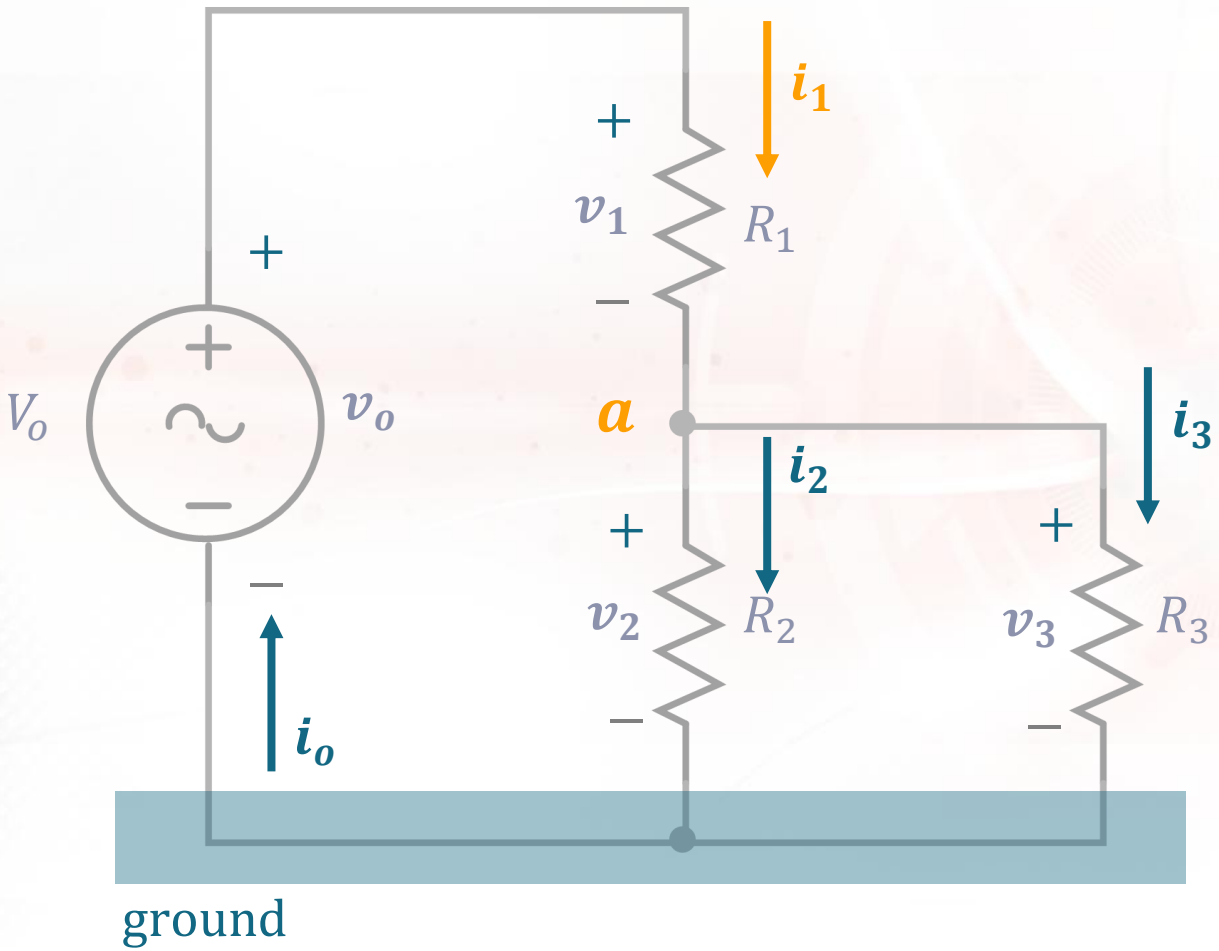


Uses **KCL** implicitly to formula $\boldsymbol{vi}$ relationship equation

KCL @ a

$$-\frac{v_1}{R_1} + i_2 + i_3 = 0$$

NODE ANALYSIS METHOD

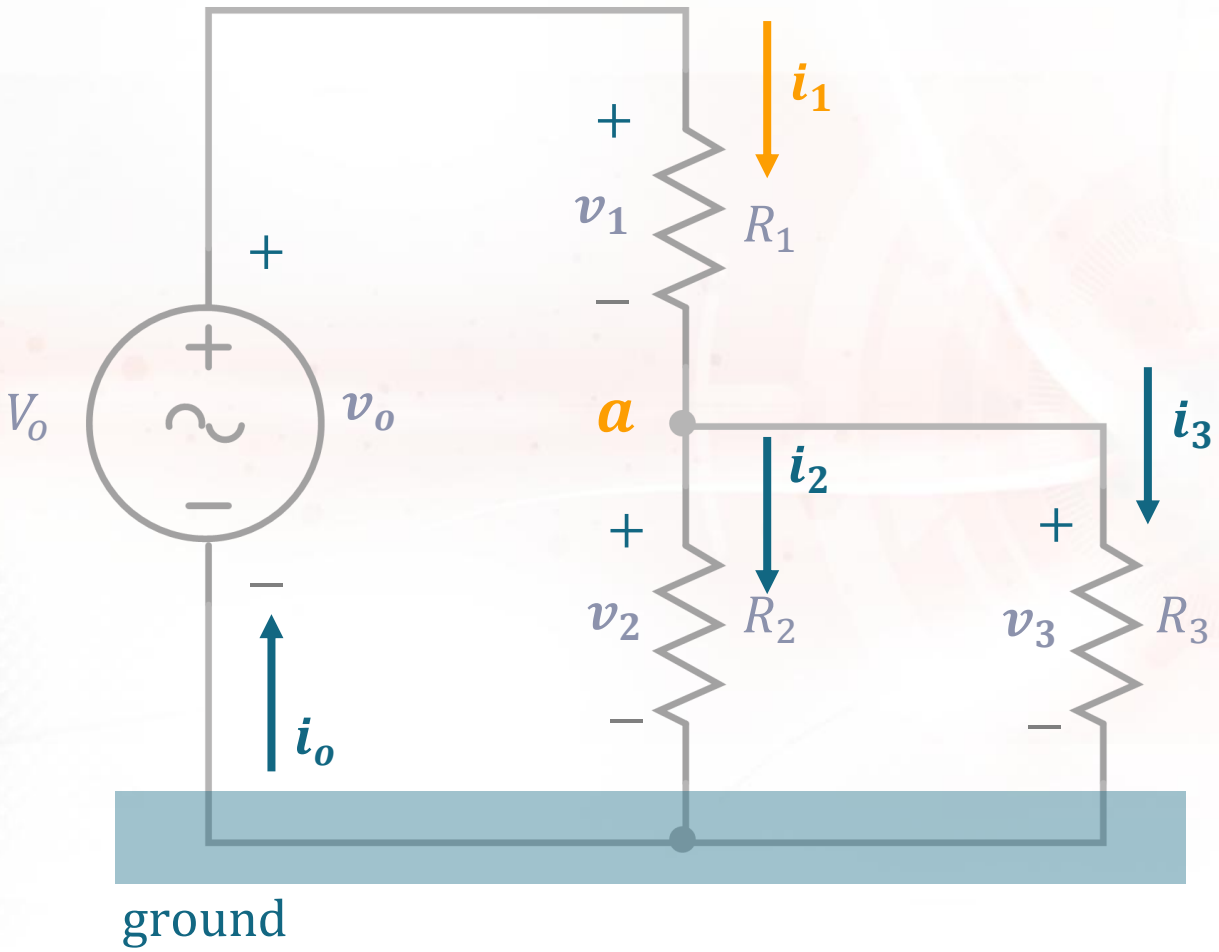


Uses **KCL** implicitly to formula $\boldsymbol{vi}$ relationship equation

KCL @ a

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + i_3 = 0$$

NODE ANALYSIS METHOD

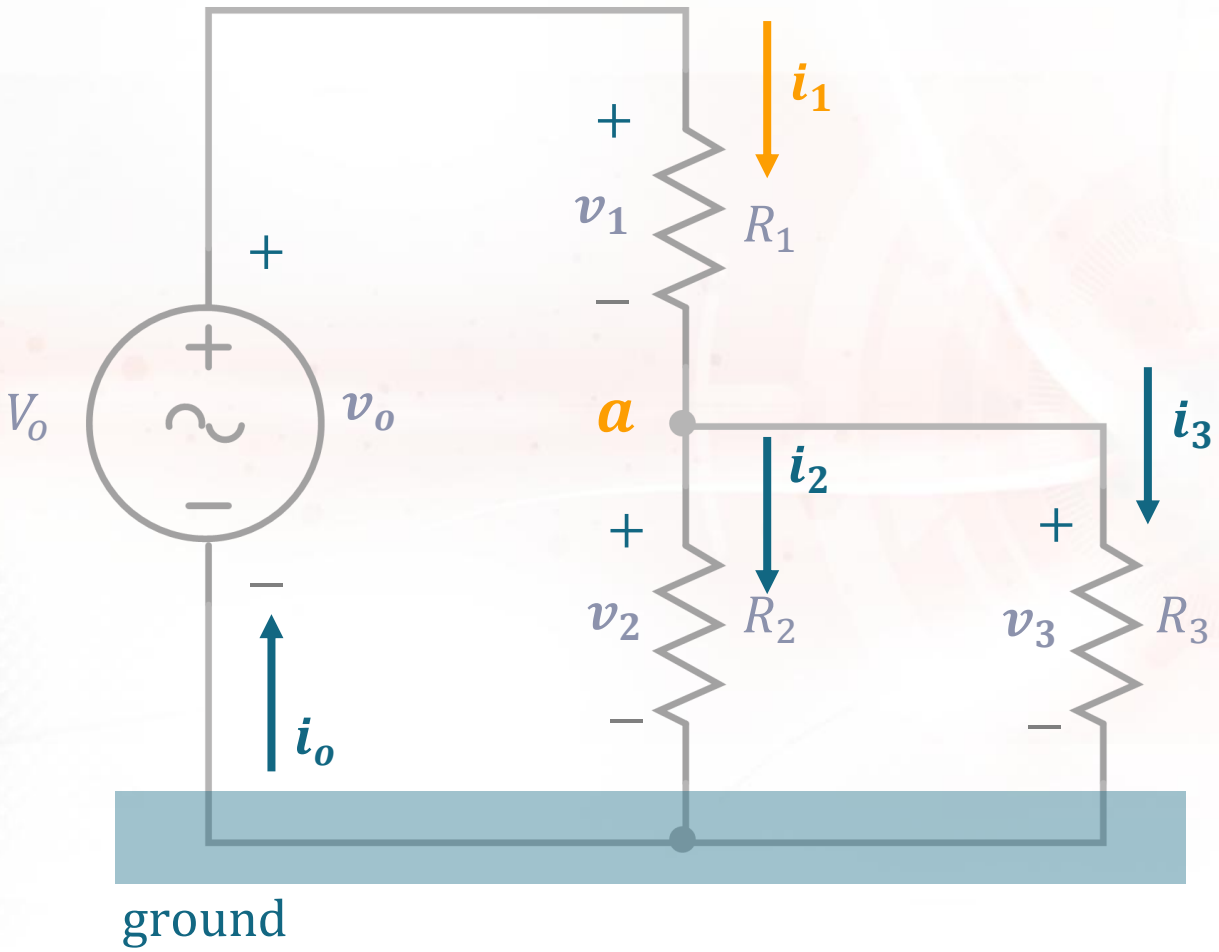


Uses **KCL** implicitly to formula $\boldsymbol{vi}$ relationship equation

KCL @ a

$$-\frac{v_1}{R_1} \quad +\frac{v_2}{R_2} \quad +\frac{v_3}{R_3} \quad = 0$$

NODE ANALYSIS METHOD

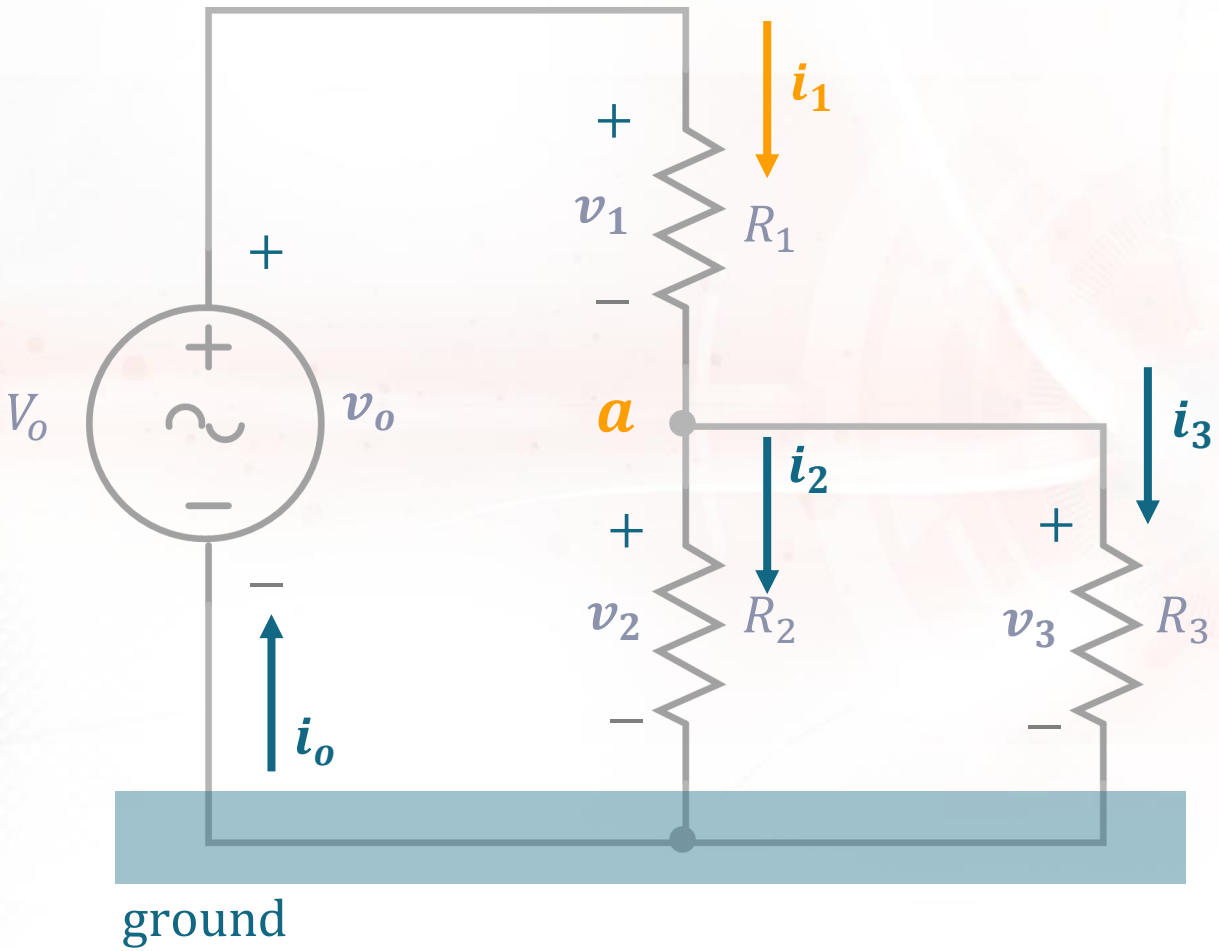


Uses **KCL** implicitly to formula ***vi*** relationship equation

KCL @ *a*

$$-\frac{v_o - v_a}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

NODE ANALYSIS METHOD

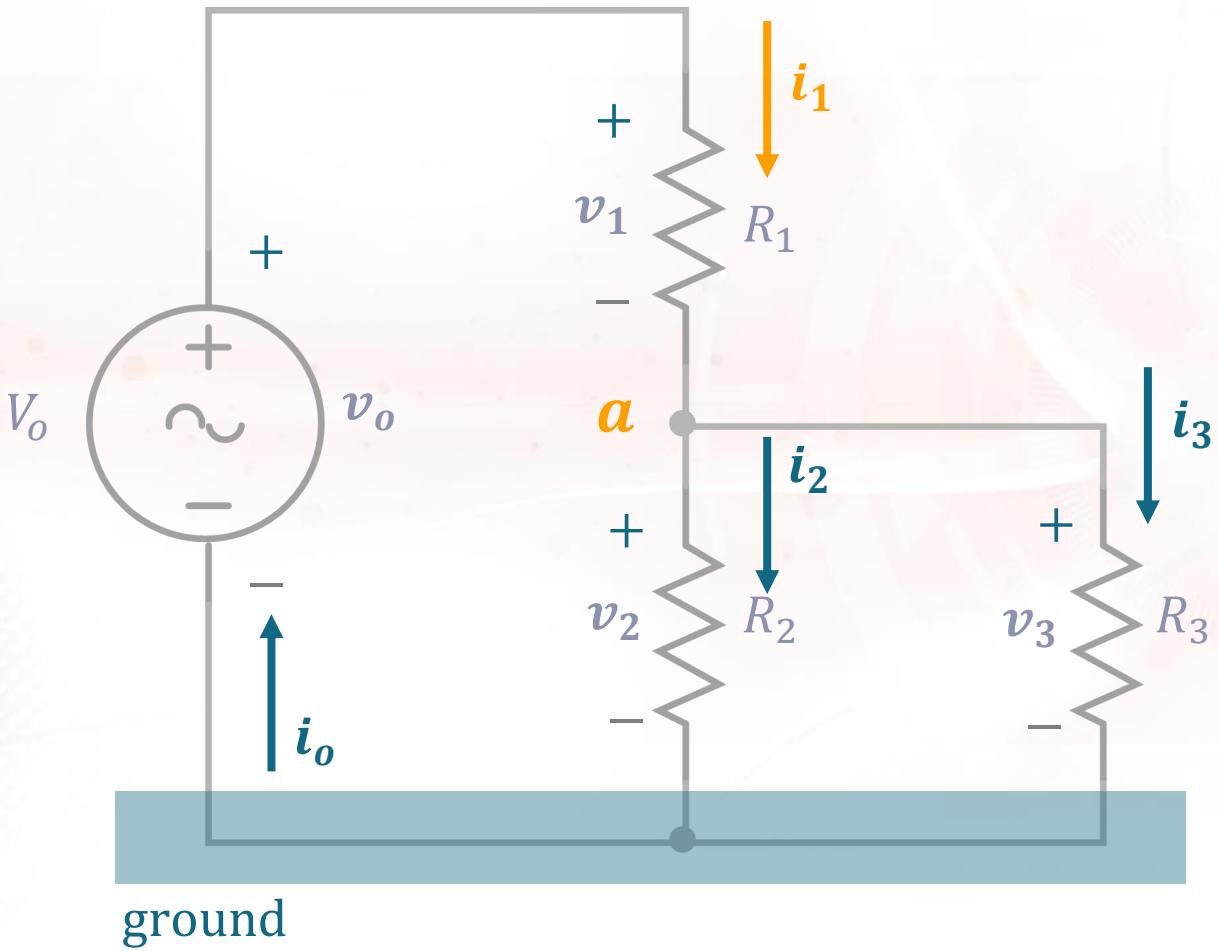


Uses **KCL** implicitly to formula $\boldsymbol{vi}$ relationship equation

KCL @ a

$$-\frac{v_o - v_a}{R_1} + \frac{v_a - 0}{R_2} + \frac{v_3}{R_3} = 0$$

NODE ANALYSIS METHOD

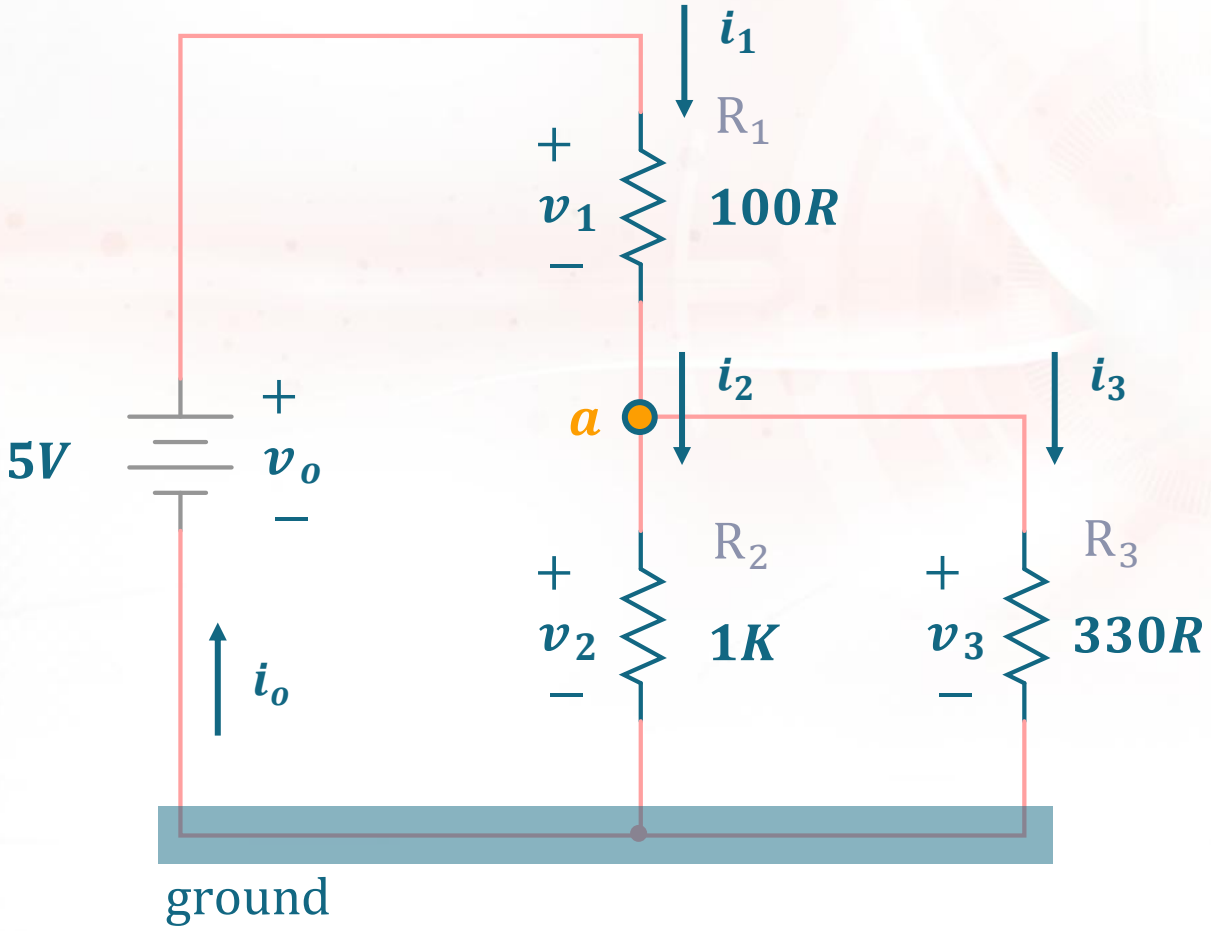


Uses **KCL** implicitly to formula ***vi*** relationship equation

KCL @ *a*

$$-\frac{v_o - v_a}{R_1} + \frac{v_a - 0}{R_2} + \frac{v_a - 0}{R_3} = 0$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



KCL @ a

$$-i_1 + i_2 + i_3 = 0$$

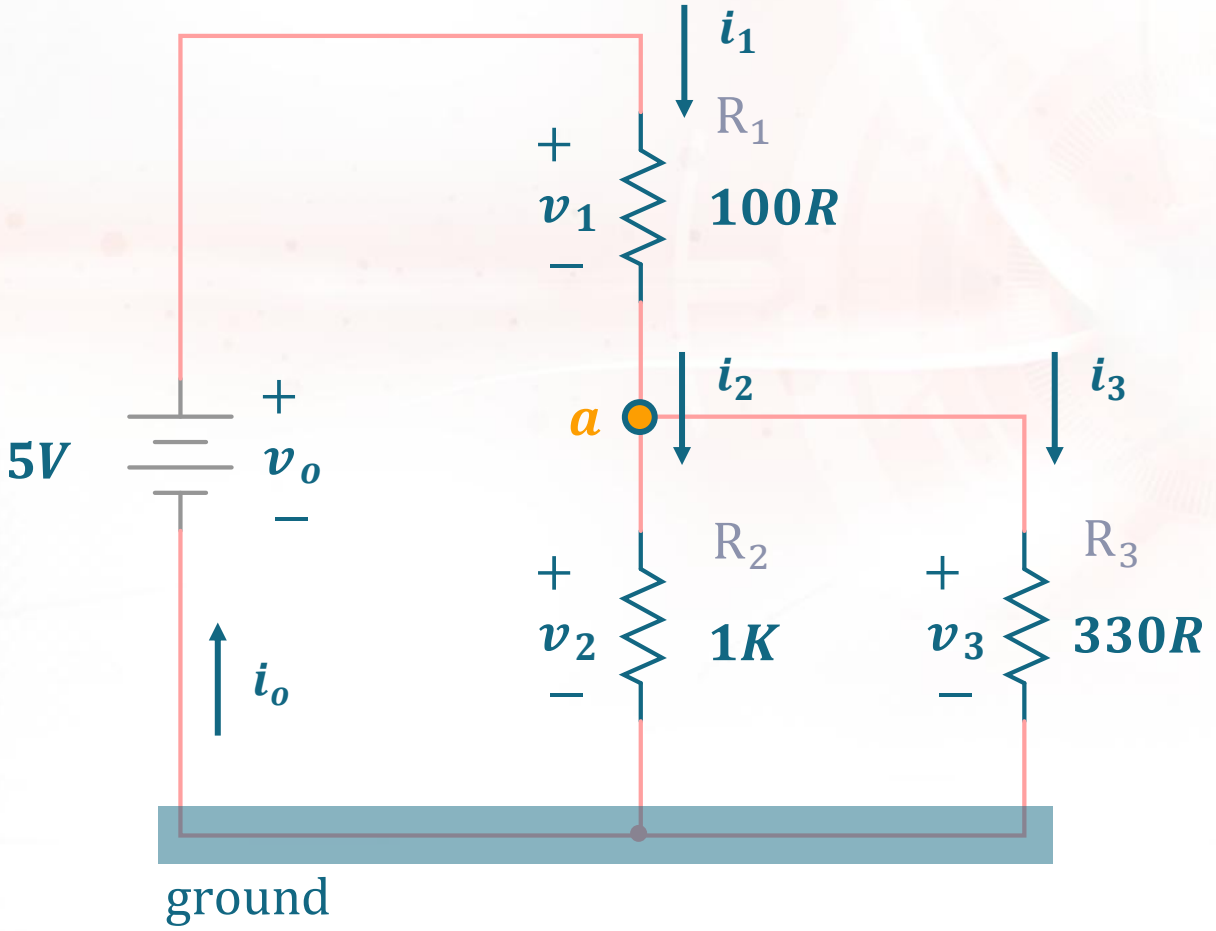
$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

$$-\frac{v_o - v_a}{R_1} + \frac{v_a - 0}{R_2} + \frac{v_a - 0}{R_3} = 0$$

$$-\frac{v_o}{R_1} + \frac{v_a}{R_1} + \frac{v_a}{R_2} + \frac{v_a}{R_3} = 0$$

$$-\frac{v_o}{R_1} + v_a\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) = 0$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$$-\frac{v_o}{R_1} + v_a\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right) = 0$$

$$v_a\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right) = \frac{v_o}{R_1}$$

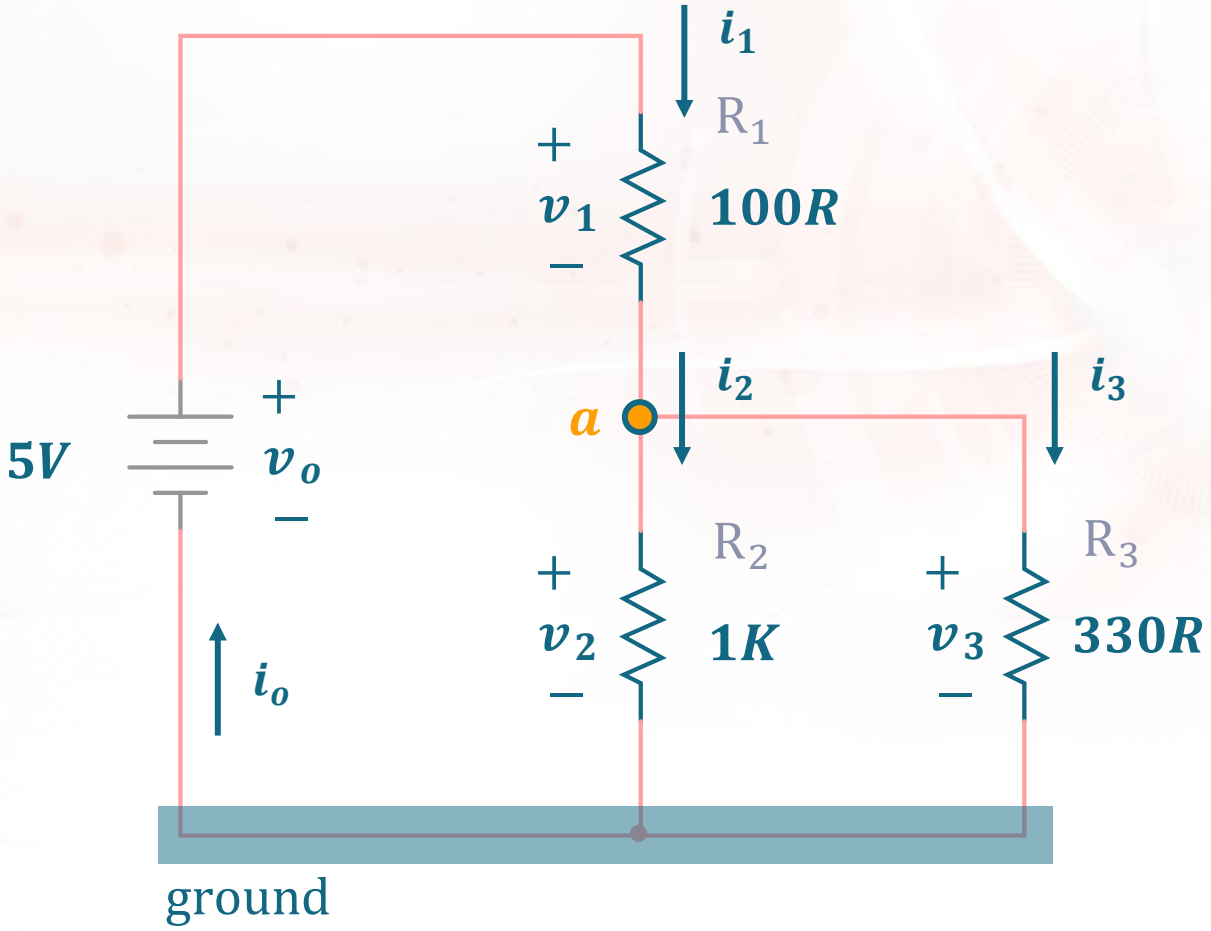
$$v_a\left(\frac{1}{100}+\frac{1}{1K}+\frac{1}{330}\right) = \frac{5}{100}$$

$$\frac{463}{33K}v_a = \frac{5}{100}$$

$$v_a = \frac{5(33K)}{100(463)}$$

$$v_a = 3.56\ V$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$$v_a = 3.56\ V$$

$$v_2 = v_a - 0$$

$$\boxed{v_2 = 3.56\ V}$$

ans

$$\boxed{v_3 = 3.56\ V}$$

ans

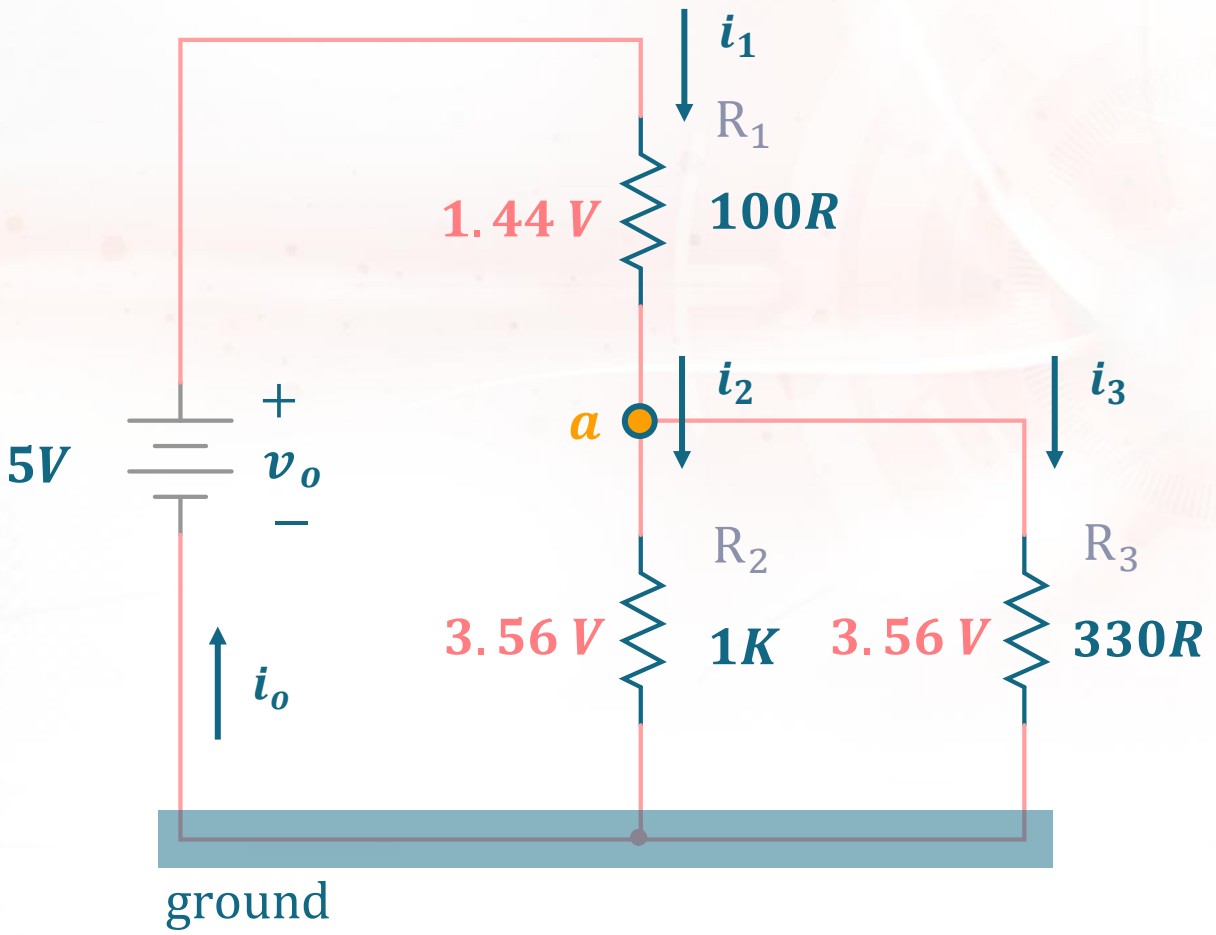
$$v_1 = v_o - v_a$$

$$v_1 = 5 - 3.56$$

$$\boxed{v_1 = 1.44\ V}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$$i_1 = \frac{v_1}{R_1}$$

$$i_1 = \frac{1.44}{100}$$

$$\boxed{i_1 = 14.4\ mA}$$

ans

$$i_2 = \frac{v_2}{R_2}$$

$$i_2 = \frac{3.56}{1000}$$

$$\boxed{i_2 = 3.56\ mA}$$

ans

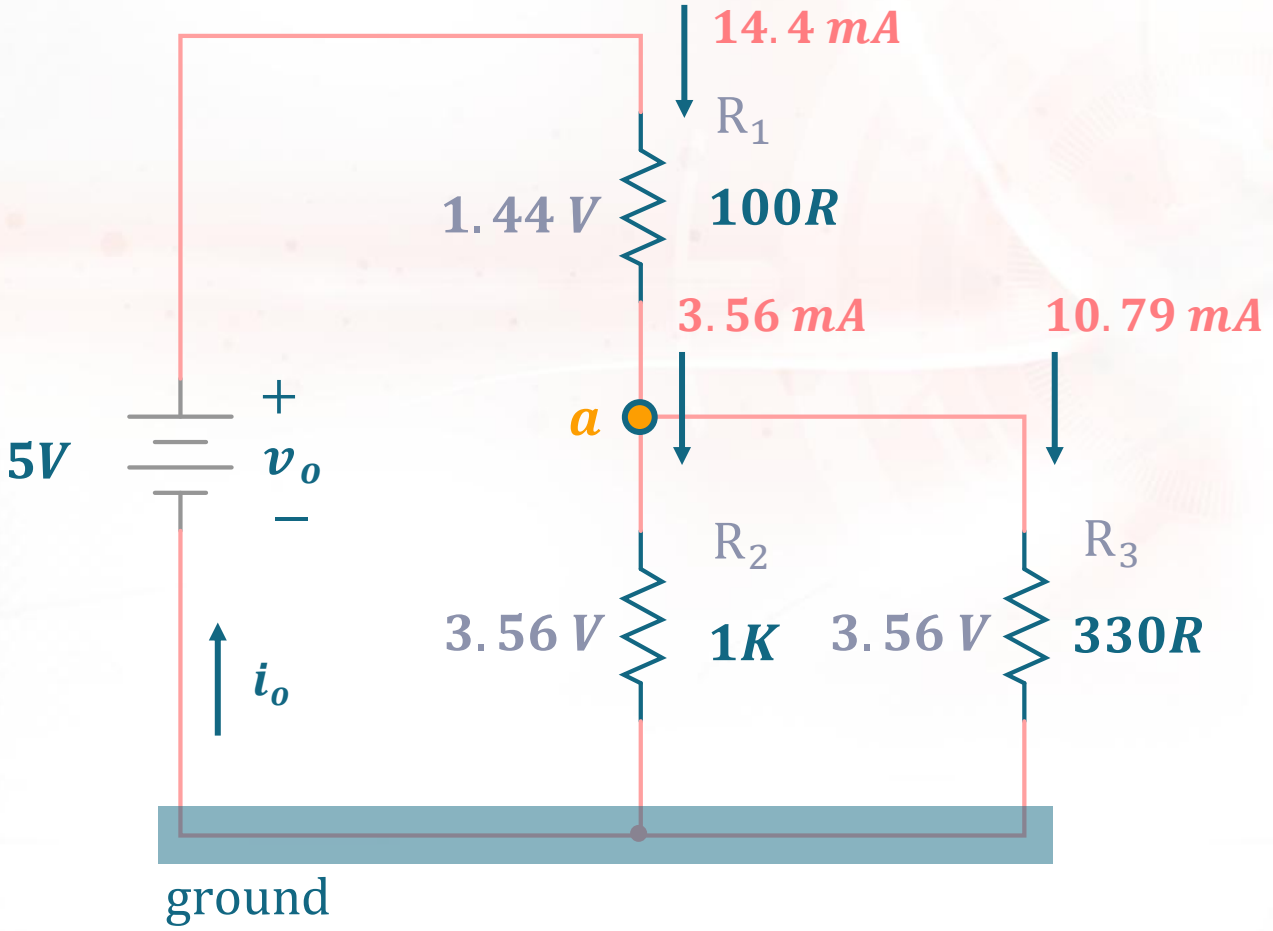
$$i_3 = \frac{v_3}{R_3}$$

$$i_3 = \frac{3.56}{330}$$

$$\boxed{i_3 = 10.79\ mA}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$$P_1 = v_1 i_1$$

$$P_1 = (1.44)14.4\ mA$$

$$\boxed{P_1 = 20.74\ mW}$$

ans

$$P_2 = v_2 i_2$$

$$P_3 = v_3 i_3$$

$$P_2 = (3.56)3.56\ mA$$

$$P_3 = (3.56)10.79\ mA$$

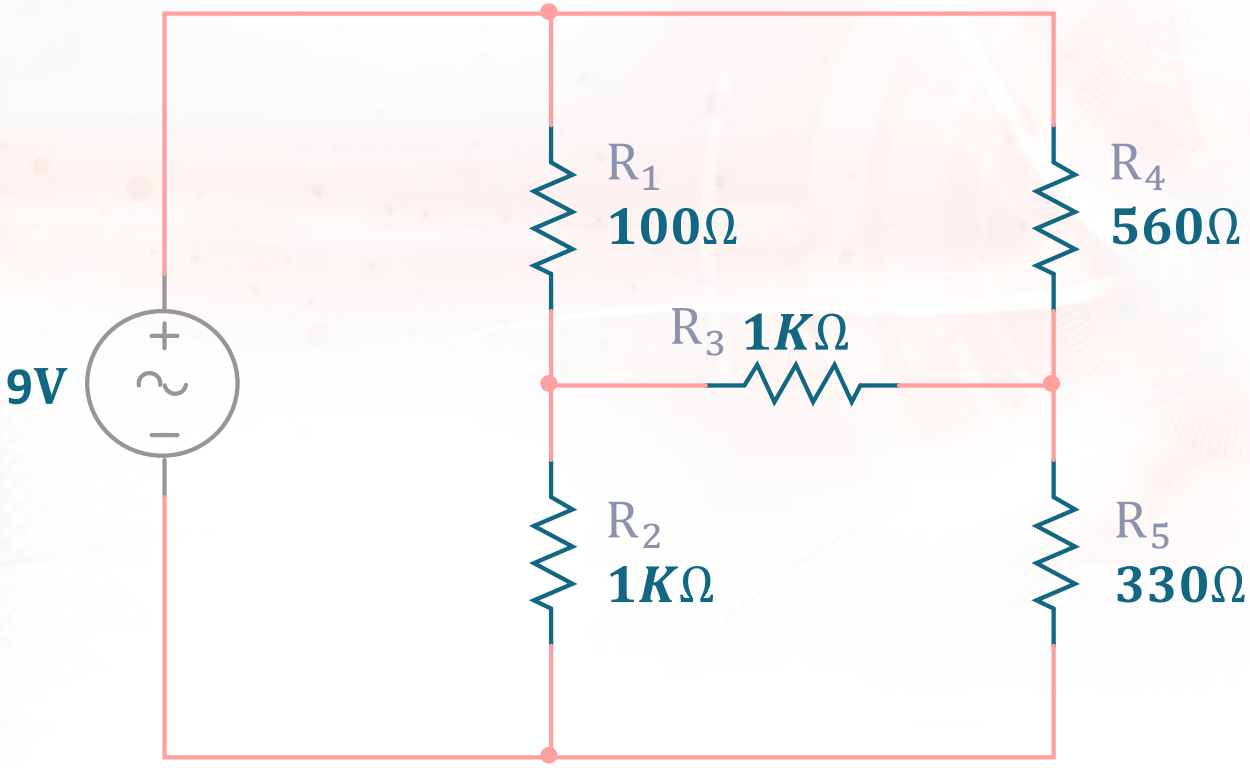
$$\boxed{P_2 = 12.64\ mW}$$

ans

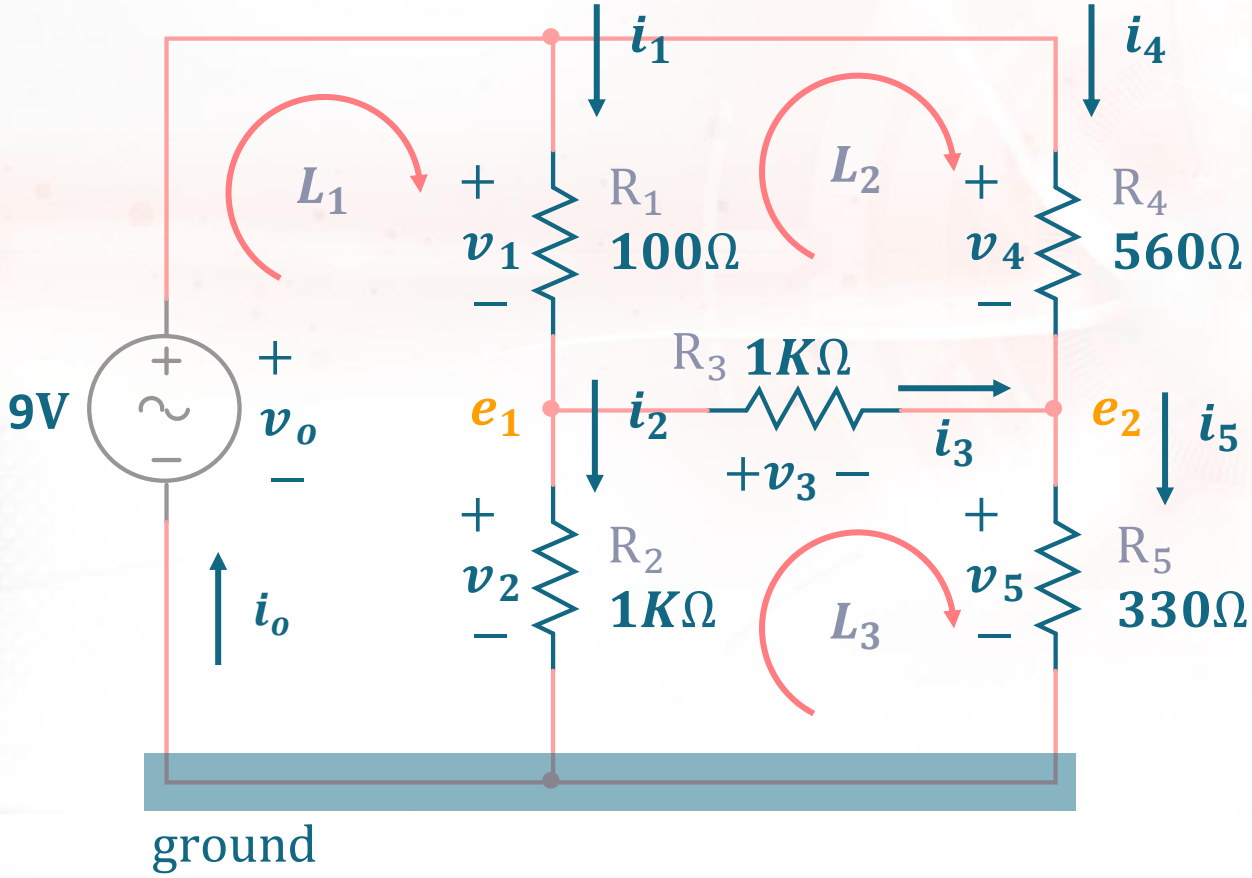
$$\boxed{P_3 = 38.41\ mW}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



KCL @ e_1

$$-i_1 + i_2 + i_3 = 0$$

$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

1 $$-\frac{v_1}{100} + \frac{v_2}{1K} + \frac{v_3}{1K} = 0$$

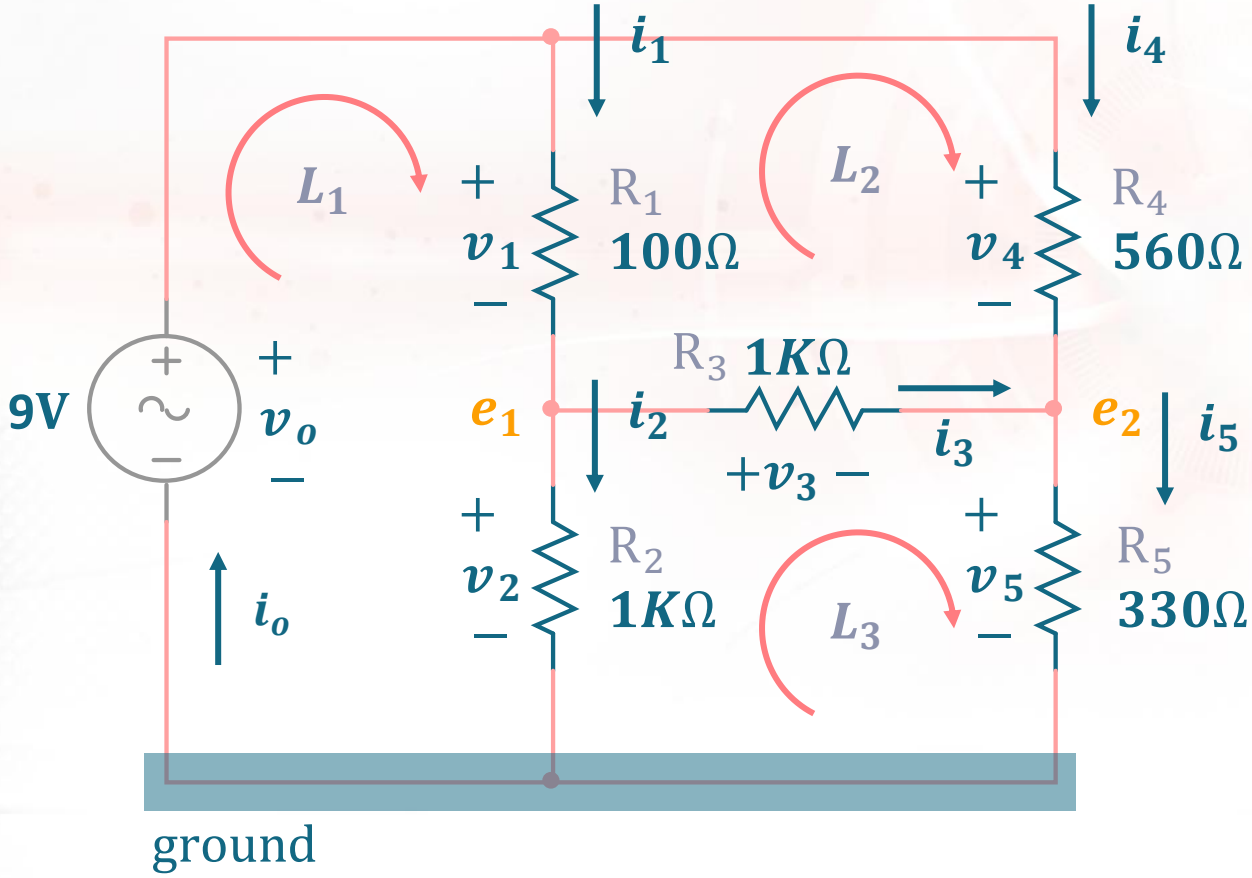
KCL @ e_2,

$$-i_3 - i_4 + i_5 = 0$$

$$-\frac{v_3}{R_3} - \frac{v_4}{R_4} + \frac{v_5}{R_5} = 0$$

2 $$-\frac{v_3}{1K} - \frac{v_4}{560} + \frac{v_5}{330} = 0$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



KVL @ L_1

$$-v_o + v_1 + v_2 = 0$$

$$-9 + v_1 + v_2 = 0$$

3 $$v_1 + v_2 = 9$$

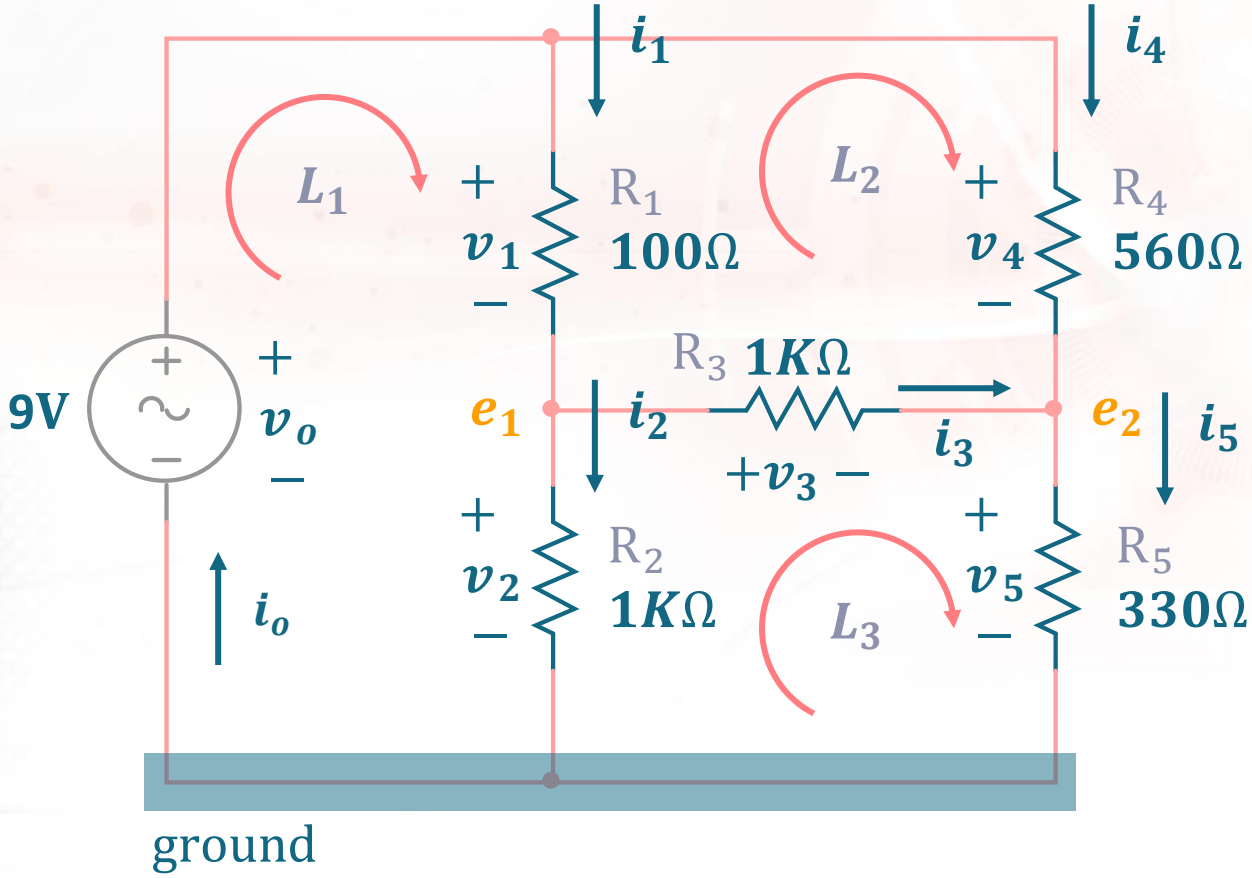
KVL @ L_2,

4 $$-v_1 + v_4 - v_3 = 0$$

KVL @ L_3,

5 $$-v_2 + v_3 + v_5 = 0$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



Linear Equations

$$-\frac{v_1}{100}+\frac{v_2}{1K}+\frac{v_3}{1K}=0$$

$$-\frac{v_3}{1K}-\frac{v_4}{560}+\frac{v_5}{330}=0$$

$$v_1+v_2=9$$

$$-v_1+v_4-v_3=0$$

$$-v_2+v_3+v_5=0$$

Augmented Matrix

$$
\begin{array}{cc}
 & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\
 & \begin{bmatrix}
-\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\
0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\
1 & 1 & 0 & 0 & 0 & 9 \\
-1 & 0 & -1 & 1 & 0 & 0 \\
0 & -1 & 1 & 0 & 1 & 0
\end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}
\end{array}
$$

Linear Equations

$$-\frac{v_1}{100} + \frac{v_2}{1K} + \frac{v_3}{1K} = 0$$

$$-\frac{v_3}{1K} - \frac{v_4}{560} + \frac{v_5}{330} = 0$$

$$v_1 + v_2 = 9$$

$$-v_1 - v_3 + v_4 = 0$$

$$-v_2 + v_3 + v_5 = 0$$

Augmented Matrix

$$
\begin{array}{c}
\begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\
\left[\begin{array}{ccccc|c}
-\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\
0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\
1 & 1 & 0 & 0 & 0 & 9 \\
-1 & 0 & -1 & 1 & 0 & 0 \\
0 & -1 & 1 & 0 & 1 & 0
\end{array}\right]
\end{array}
\begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}
$$

$$
\begin{array}{c}
\begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\
\left[\begin{array}{ccccc|c}
-\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\
0 & -1 & 1 & 0 & 1 & 0 \\
0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\
-1 & 0 & -1 & 1 & 0 & 0 \\
1 & 1 & 0 & 0 & 0 & 9
\end{array}\right]
\end{array}
\begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}
$$

Augmented Matrix

$$\begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 1 & -1 & 1 & 0 & 9 \\ 1 & 1 & 0 & 0 & 0 & 9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

$$\begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

$$r_4 \leftarrow r_4 + r_5$$

Augmented Matrix

$$\begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 1 & -1 & 1 & 0 & 9 \\ 1 & 1 & 0 & 0 & 0 & 9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

$$\begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 1 & 1 & 0 & 0 & 0 & 9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

$r_4 \leftarrow r_4 + r_2$

Augmented Matrix

$$\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ & \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 1.1 & 0.1 & 0 & 0 & 9 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array} \end{array}$$

$$\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ & \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 1 & 1 & 0 & 0 & 0 & 9 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array} \end{array}$$

$$r_5 \leftarrow 100r_1 + r_5$$

Augmented Matrix

$$\begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 1.1 & 0.1 & 0 & 0 & 9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

$$\begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 1.2 & 0 & 1.1 & 9 \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

$r_5 \leftarrow 1.1r_2 + r_5$

Augmented Matrix

$$\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ & \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 0 & -\frac{1.2}{560} & \frac{521}{110K} & \frac{9}{1K} \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array} \end{array}$$

$$\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ & \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 1.2 & 0 & 1.1 & 9 \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array} \end{array}$$

$$r_5 \leftarrow 1.2r_3 + \frac{1}{1K}r_5$$

Augmented Matrix

$$\begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 0 & -\frac{1.2}{560} & \frac{521}{110K} & \frac{9}{1K} \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

$$\begin{array}{c} \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 0 & 0 & \frac{5297}{770K} & \frac{99}{3.5K} \end{bmatrix} \end{array} \begin{array}{c} \\ r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

$$r_5 \leftarrow \frac{1.2}{560} r_4 + r_5$$

From equation 5

$$\frac{5297}{770K}v_5 = \frac{99}{3.5K}$$

$$v_5 = \frac{99(770K)}{3.5K(5297)}$$

$$\boxed{v_5 = 4.11\,V}$$

ans

From equation 4

$$v_4 + v_5 = 9$$

$$v_4 + 4.11 = 9$$

$$v_4 = 9 - 4.11$$

$$\boxed{v_4 = 4.89\,V}$$

ans

$$\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ & \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 0 & 0 & \frac{5297}{770K} & \frac{99}{3.5K} \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array} \end{array}$$

From equation 3

$$-\frac{v_3}{1K}-\frac{v_4}{560}+\frac{v_5}{330}=0$$

$$-\frac{v_3}{1K}-\frac{4.89}{560}+\frac{4.11}{330}=0$$

$$-\frac{v_3}{1K}+\frac{2293}{616K}=0$$

$$\frac{v_3}{1K}=\frac{2293}{616K}$$

$$\boxed{v_3=3.72\,V}$$

ans

$$v_5=4.11\,V$$

$$v_4=4.89\,V$$

$$\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ & \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 0 & 0 & \frac{5297}{770K} & \frac{99}{3.5K} \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array} \end{array}$$

From equation 2

$$-v_2 + v_3 + v_5 = 0$$

$$-v_2 + 3.72 + 4.11 = 0$$

$$-v_2 + 7.83 = 0$$

$$\boxed{v_2 = 7.83\ V}$$

ans

$$v_5 = 4.11\ V$$

$$v_4 = 4.89\ V$$

$$v_3 = 3.72\ V$$

$$\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ & \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 0 & 0 & \frac{5297}{770K} & \frac{99}{3.5K} \end{bmatrix} \end{array} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array}$$

From equation 1

$$-\frac{v_1}{100}+\frac{v_2}{1K}+\frac{v_3}{1K}=0$$

$$-\frac{v_1}{100}+\frac{7.83}{1K}+\frac{3.72}{1K}=0$$

$$-\frac{v_1}{100}+\frac{231}{20K}=0$$

$$\frac{v_1}{100}=\frac{231}{20K}$$

$$v_1=\frac{231(100)}{20K}$$

$$\boxed{v_1=1.16\,V}$$

ans

$v_5 = 4.11\,V$

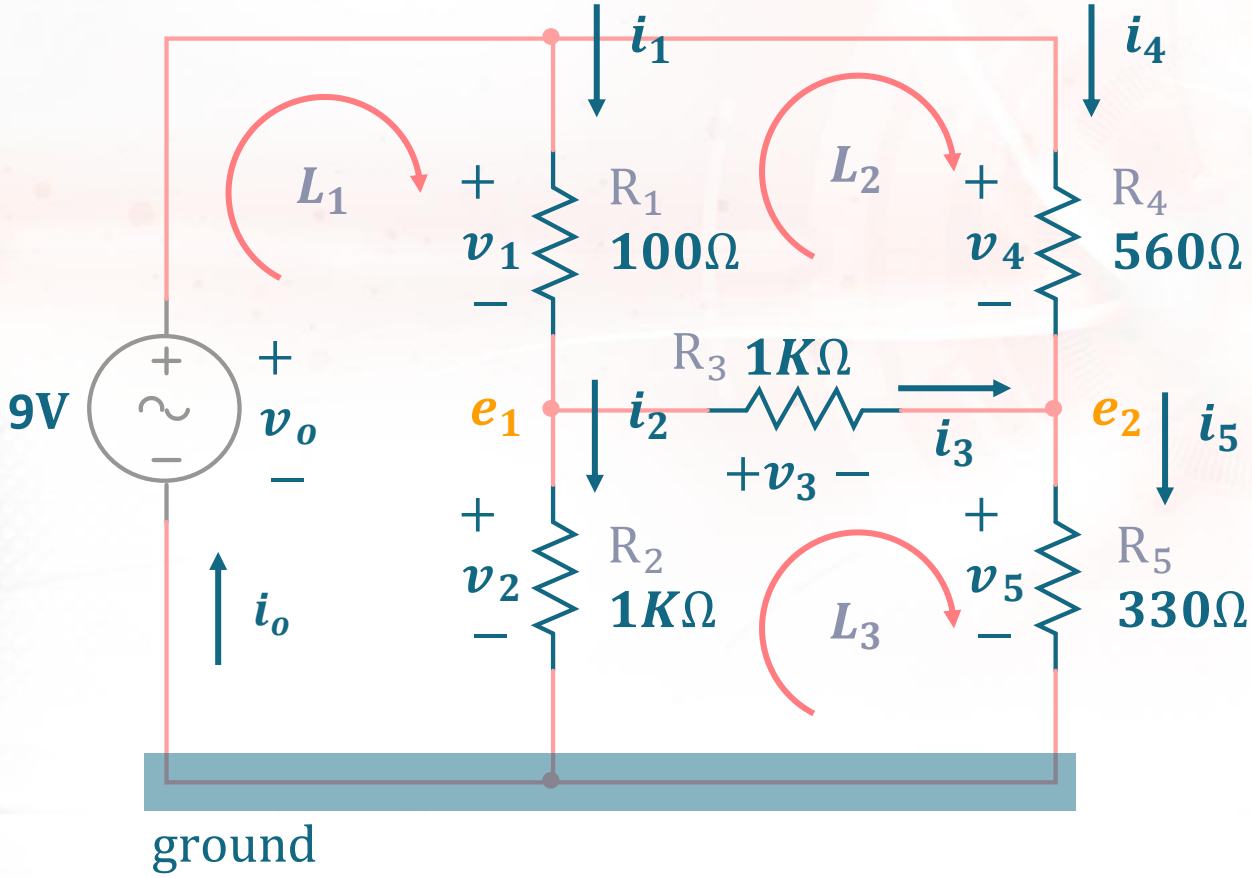
$v_4 = 4.89\,V$

$v_3 = 3.72\,V$

$v_2 = 7.83\,V$

$$\begin{array}{cc} & \begin{array}{cccccc} v_1 & v_2 & v_3 & v_4 & v_5 & \end{array} \\ & \begin{bmatrix} -\frac{1}{100} & \frac{1}{1K} & \frac{1}{1K} & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{1K} & -\frac{1}{560} & \frac{1}{330} & 0 \\ 0 & 0 & 0 & 1 & 1 & 9 \\ 0 & 0 & 0 & 0 & \frac{5297}{770K} & \frac{99}{3.5K} \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \end{array} \end{array}$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$v_1 = 1.16\ V$

$v_2 = 7.83\ V$

$v_3 = 3.72\ V$

$v_4 = 4.89\ V$

$v_5 = 4.11\ V$

$$i_1 = \frac{v_1}{R_1}$$

$$i_2 = \frac{v_2}{R_2}$$

$$i_1 = \frac{1.16}{100}$$

$$i_2 = \frac{7.83}{1K}$$

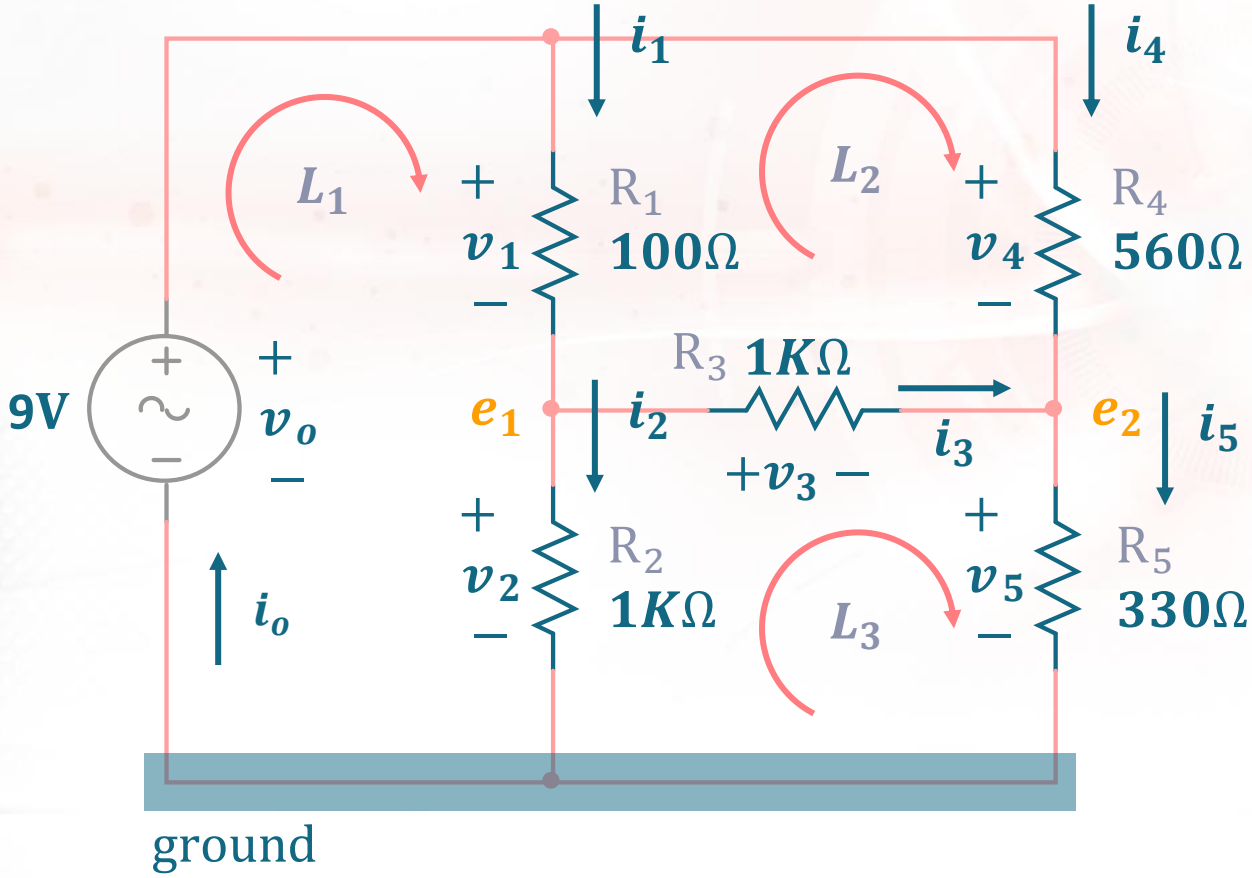
$$\boxed{i_1 = 11.6\ mA}$$

ans

$$\boxed{i_2 = 7.83\ mA}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$v_1 = 1.16\ V$

$v_2 = 7.83\ V$

$v_3 = 3.72\ V$

$v_4 = 4.89\ V$

$v_5 = 4.11\ V$

$i_1 = 11.6\ mA$

$i_2 = 7.83\ mA$

$$i_3 = \frac{v_3}{R_3}$$

$$i_3 = \frac{3.72}{1K}$$

$$\boxed{i_3 = 3.72\ mA}$$

ans

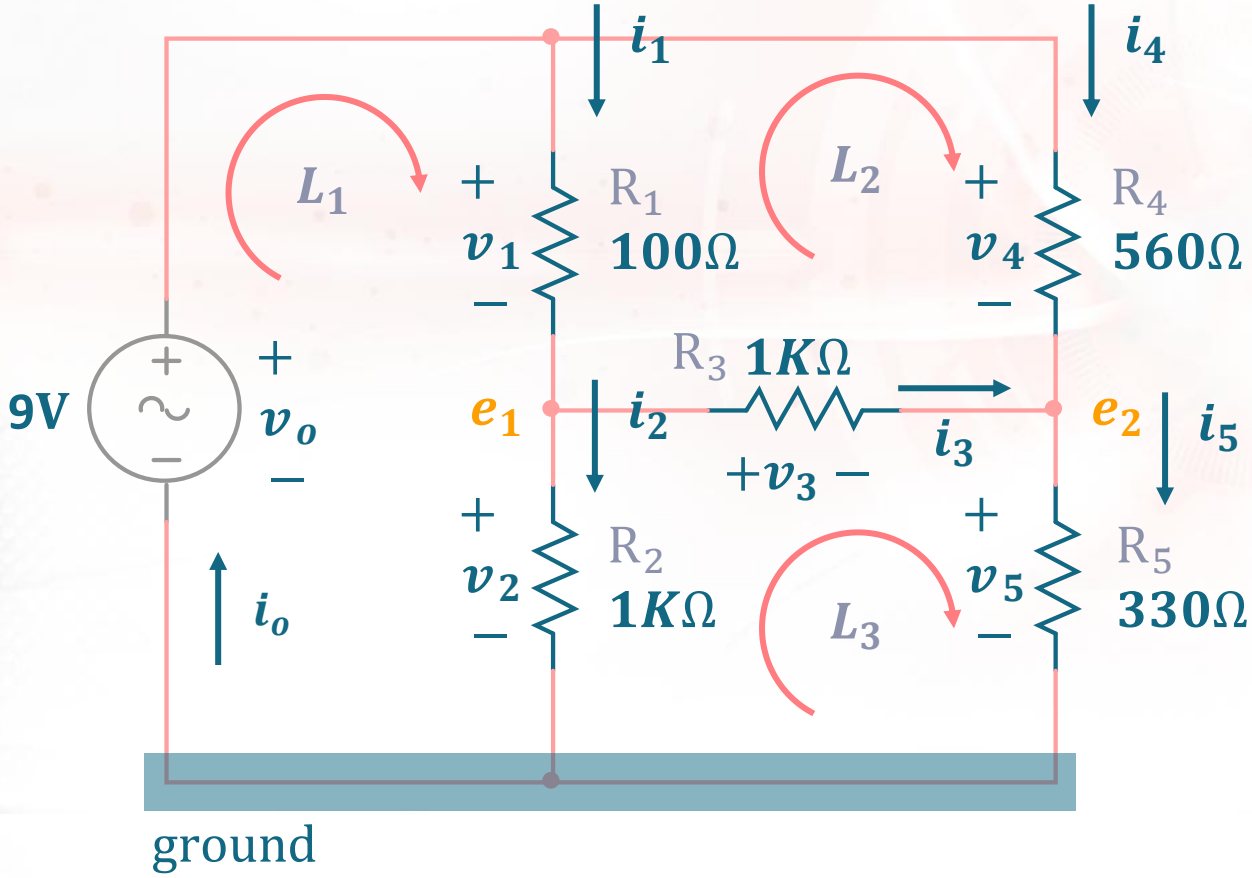
$$i_4 = \frac{v_4}{R_4}$$

$$i_4 = \frac{4.89}{560}$$

$$\boxed{i_4 = 8.73\ mA}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$v_1 = 1.16\ V$

$v_2 = 7.83\ V$

$v_3 = 3.72\ V$

$v_4 = 4.89\ V$

$v_5 = 4.11\ V$

$i_1 = 11.6\ mA$

$i_2 = 7.83\ mA$

$i_3 = 3.72\ mA$

$i_4 = 8.73\ mA$

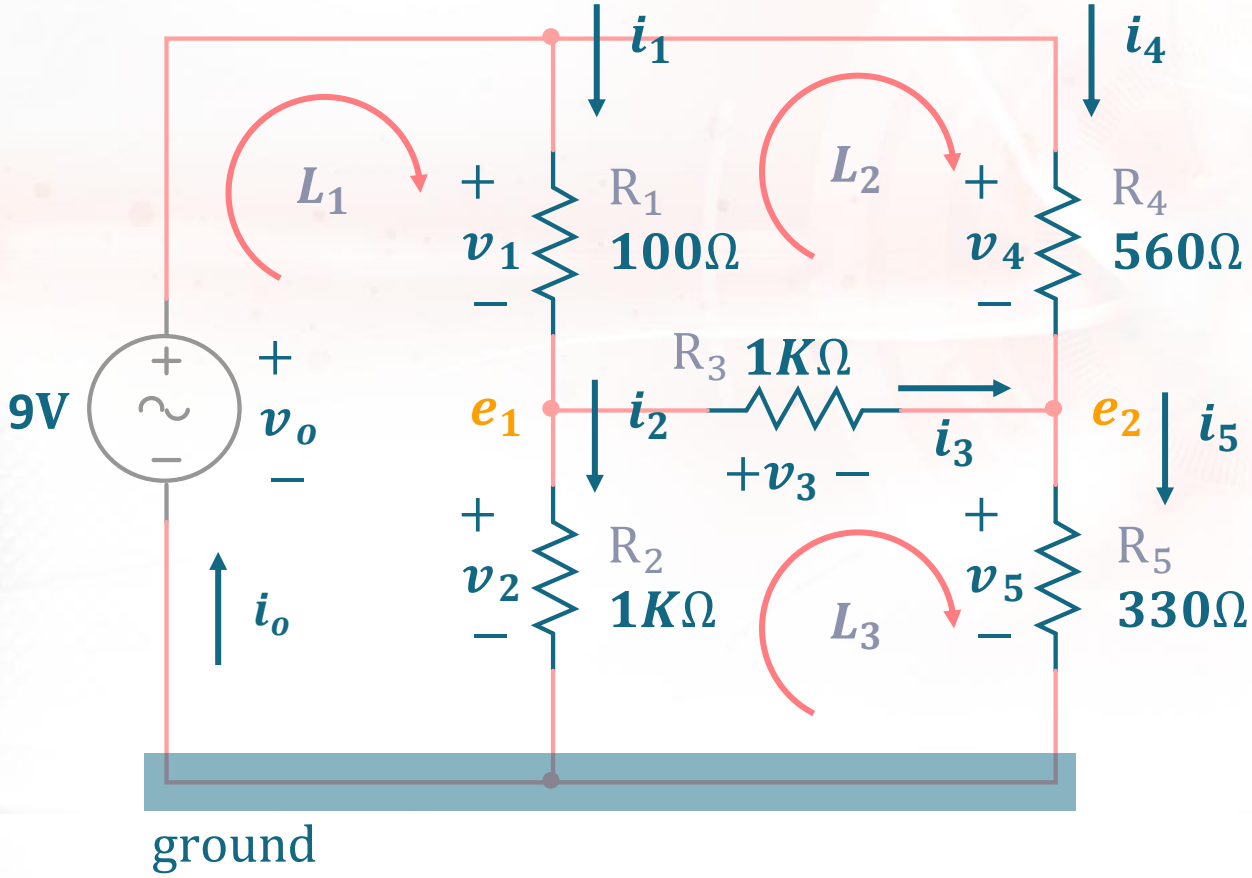
$$i_5 = \frac{v_5}{R_5}$$

$$i_5 = \frac{4.11}{330}$$

$$\boxed{i_5 = 12.45\ mA}$$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$v_1 = 1.16\ V$

$v_2 = 7.83\ V$

$v_3 = 3.72\ V$

$v_4 = 4.89\ V$

$v_5 = 4.11\ V$

$i_1 = 11.6\ mA$

$i_2 = 7.83\ mA$

$i_3 = 3.72\ mA$

$i_4 = 8.73\ mA$

$i_5 = 12.45\ mA$

$P_1 = v_1 i_1$

$P_2 = v_2 i_2$

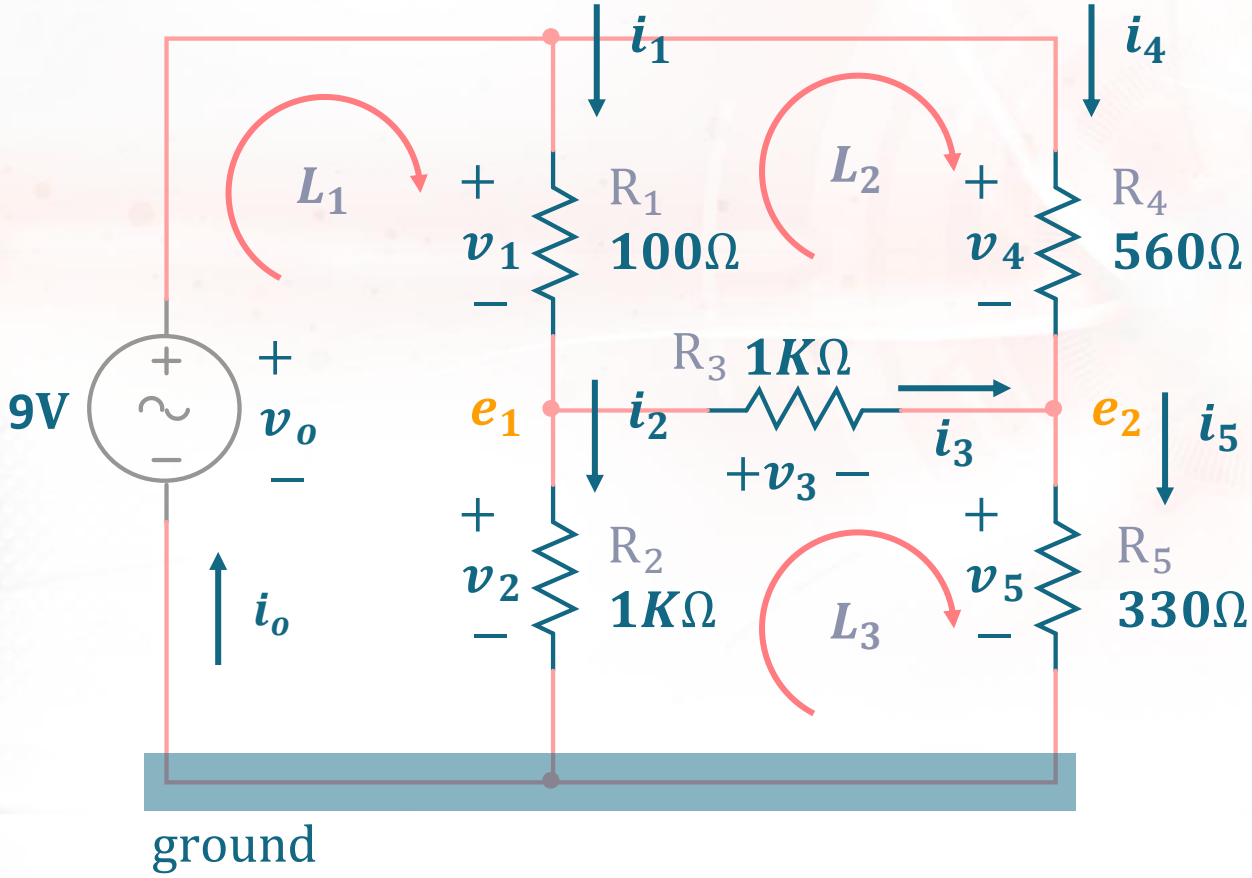
$P_1 = 1.16(11.6\ mA)$

$P_2 = 7.83(7.83\ mA)$

$\boxed{P_1 = 13.46\ mW}$ *ans*

$\boxed{P_2 = 61.31\ mW}$ *ans*

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$v_1 = 1.16\ V$

$v_2 = 7.83\ V$

$v_3 = 3.72\ V$

$v_4 = 4.89\ V$

$v_5 = 4.11\ V$

$i_1 = 11.6\ mA$

$i_2 = 7.83\ mA$

$i_3 = 3.72\ mA$

$i_4 = 8.73\ mA$

$i_5 = 12.45\ mA$

$P_3 = v_3 i_3$

$P_3 = 3.72(3.72\ mA)$

$\boxed{P_3 = 13.84\ mW}$

ans

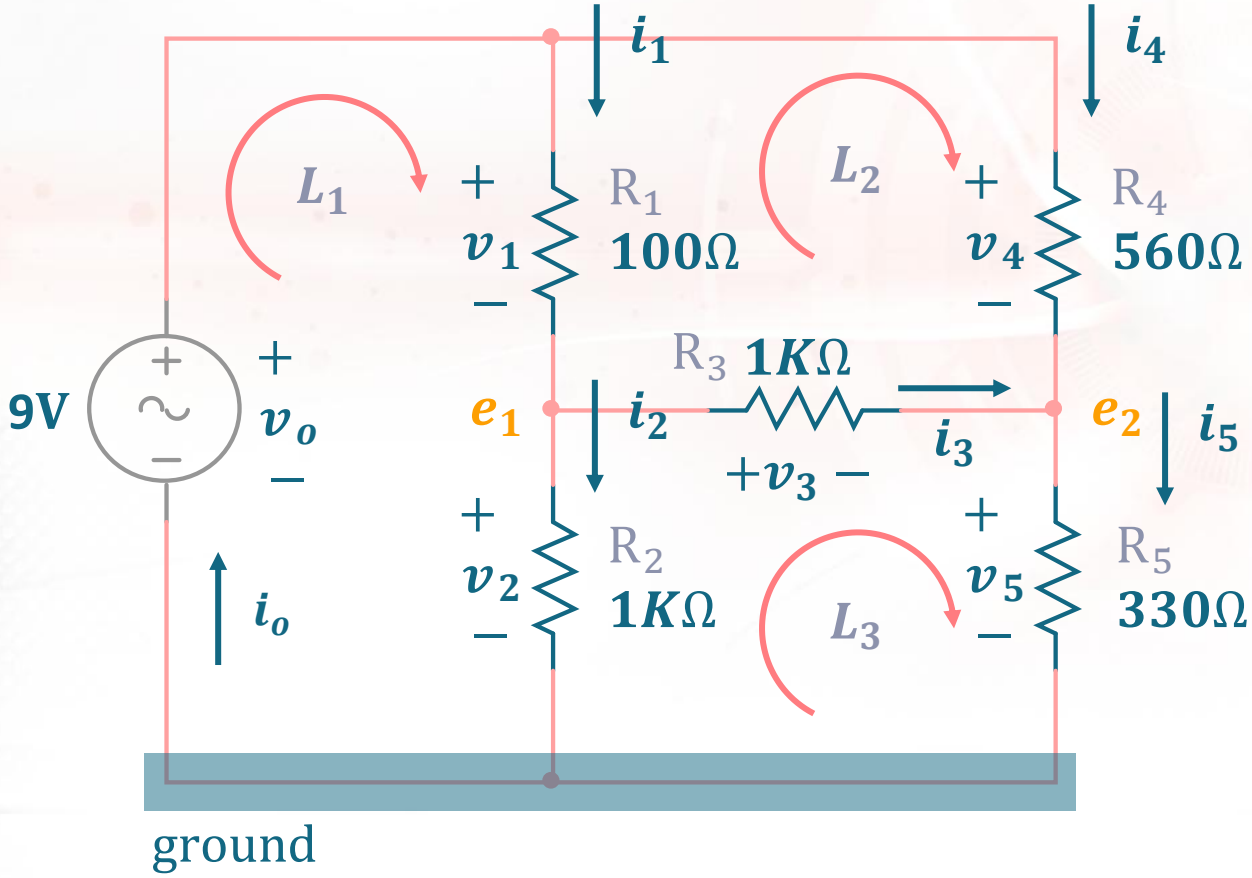
$P_4 = v_4 i_4$

$P_4 = 4.89(8.73\ mA)$

$\boxed{P_4 = 42.69\ mW}$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **KCL/KVL** method.



$v_1 = 1.16\ V$

$v_2 = 7.83\ V$

$v_3 = 3.72\ V$

$v_4 = 4.89\ V$

$v_5 = 4.11\ V$

$i_1 = 11.6\ mA$

$i_2 = 7.83\ mA$

$i_3 = 3.72\ mA$

$i_4 = 8.73\ mA$

$i_5 = 12.45\ mA$

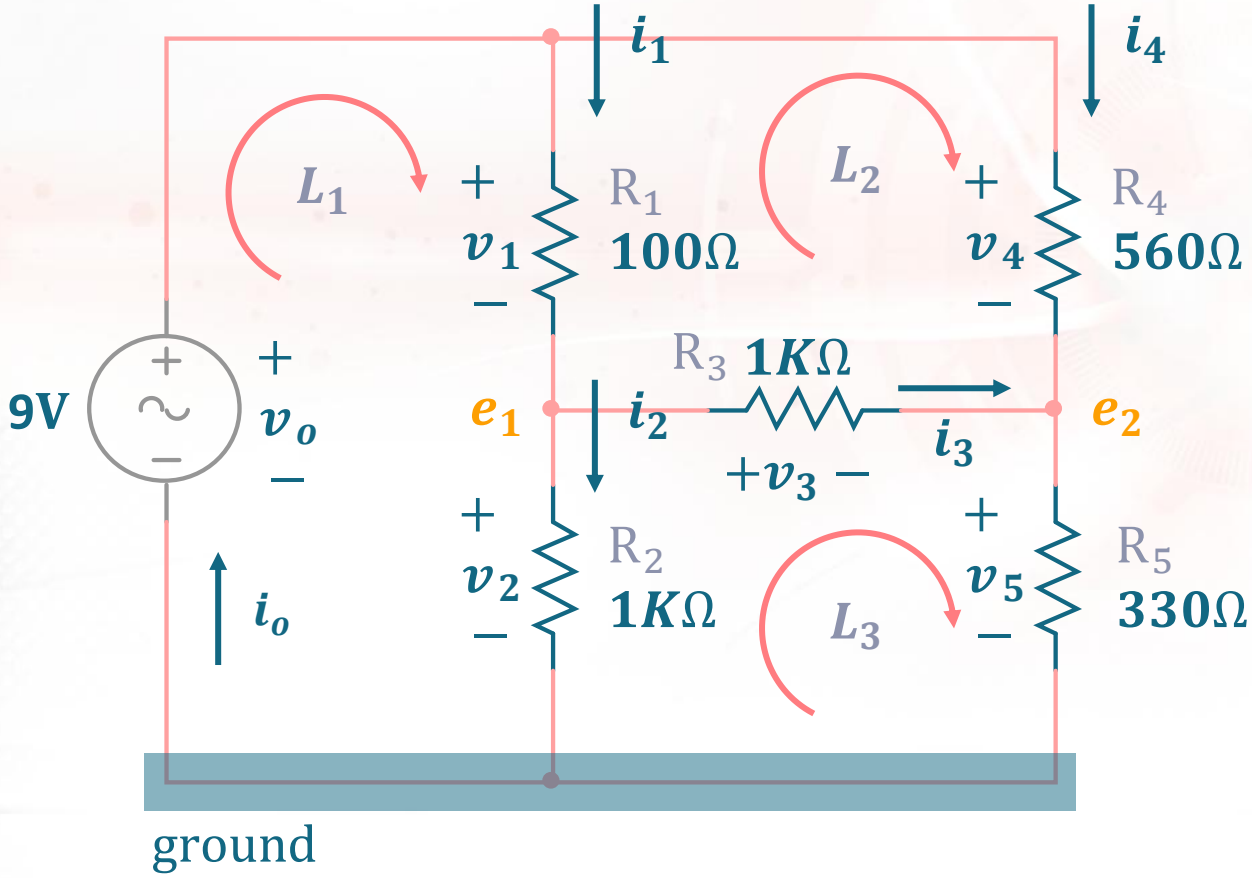
$P_5 = v_5 i_5$

$P_5 = 4.11(12.45\ mA)$

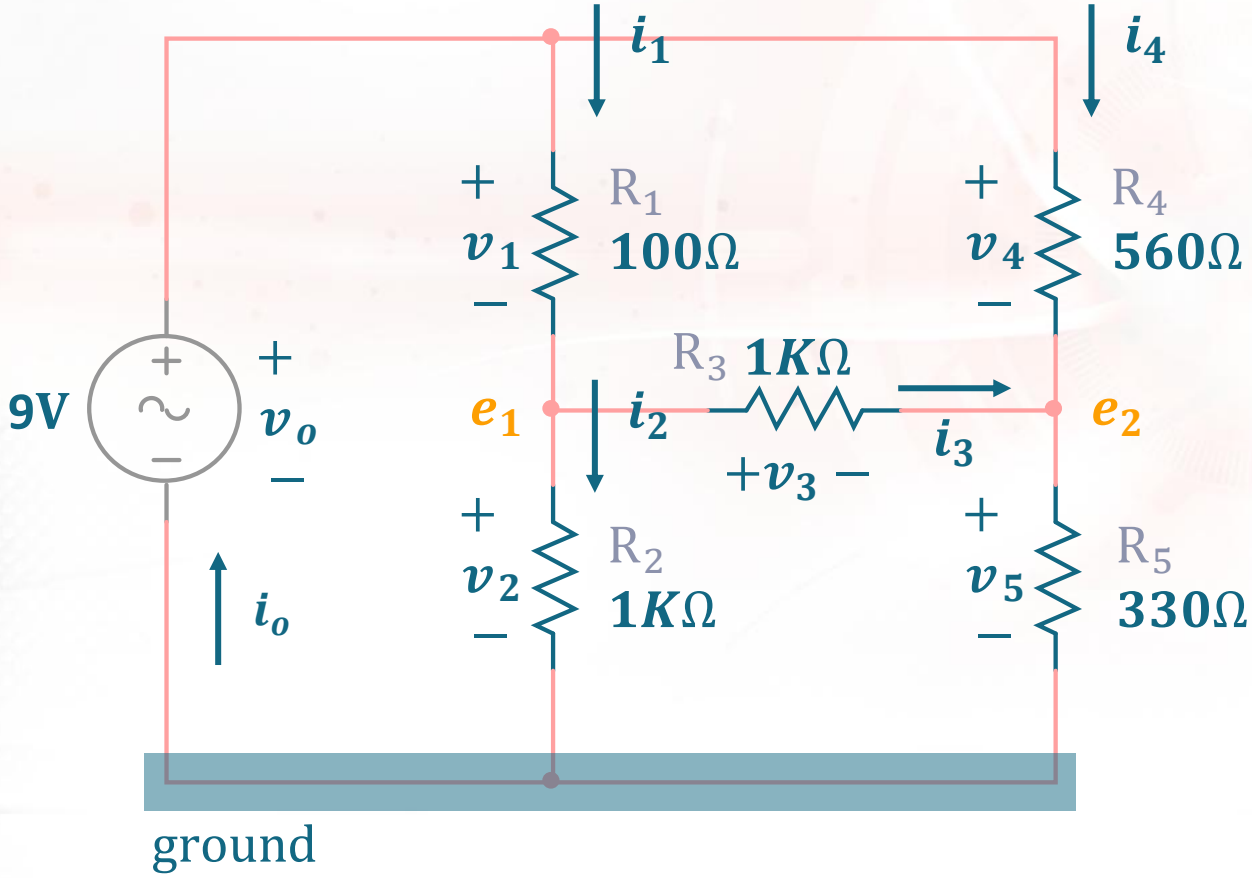
$P_5 = 51.17\ mW$

ans

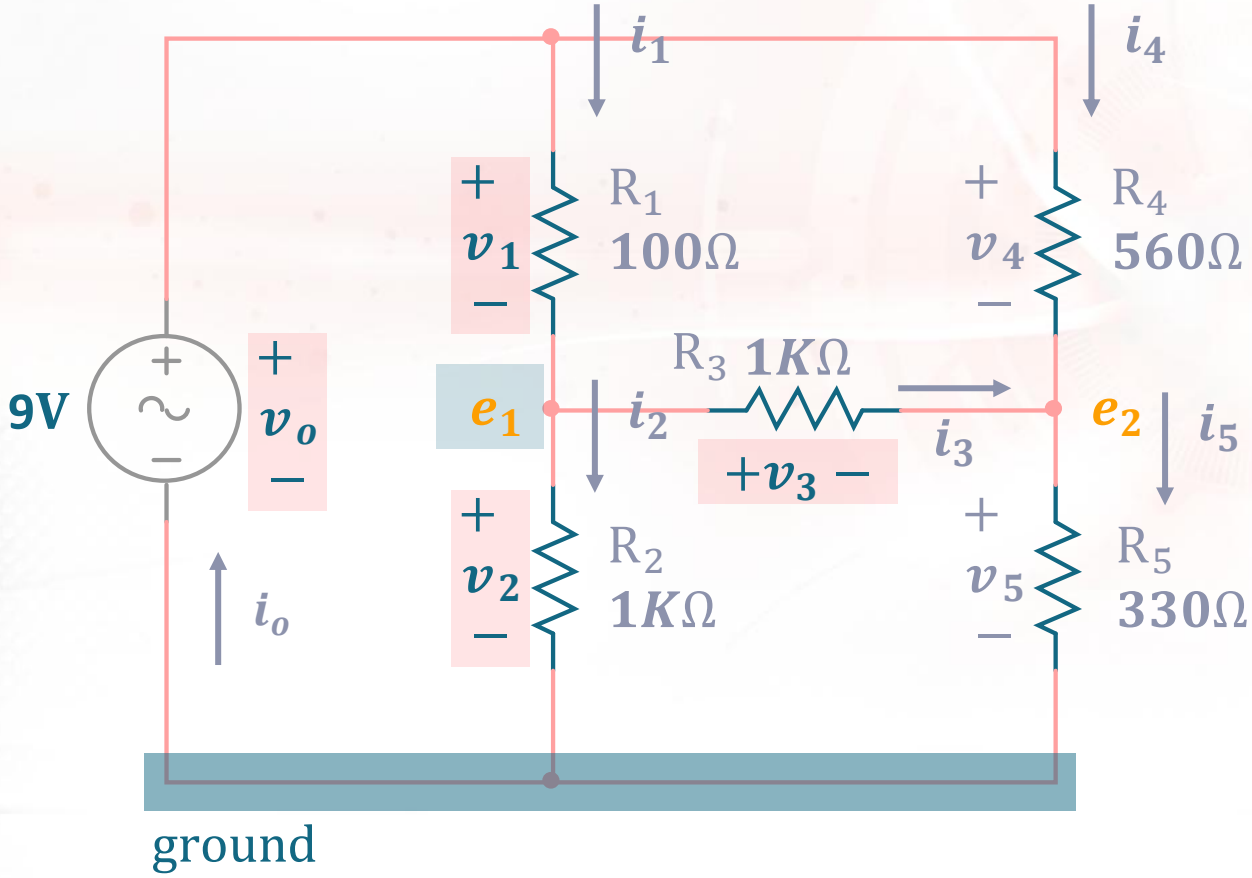
Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



KCL @ e_1

$$-i_1 + i_2 + i_3 = 0$$

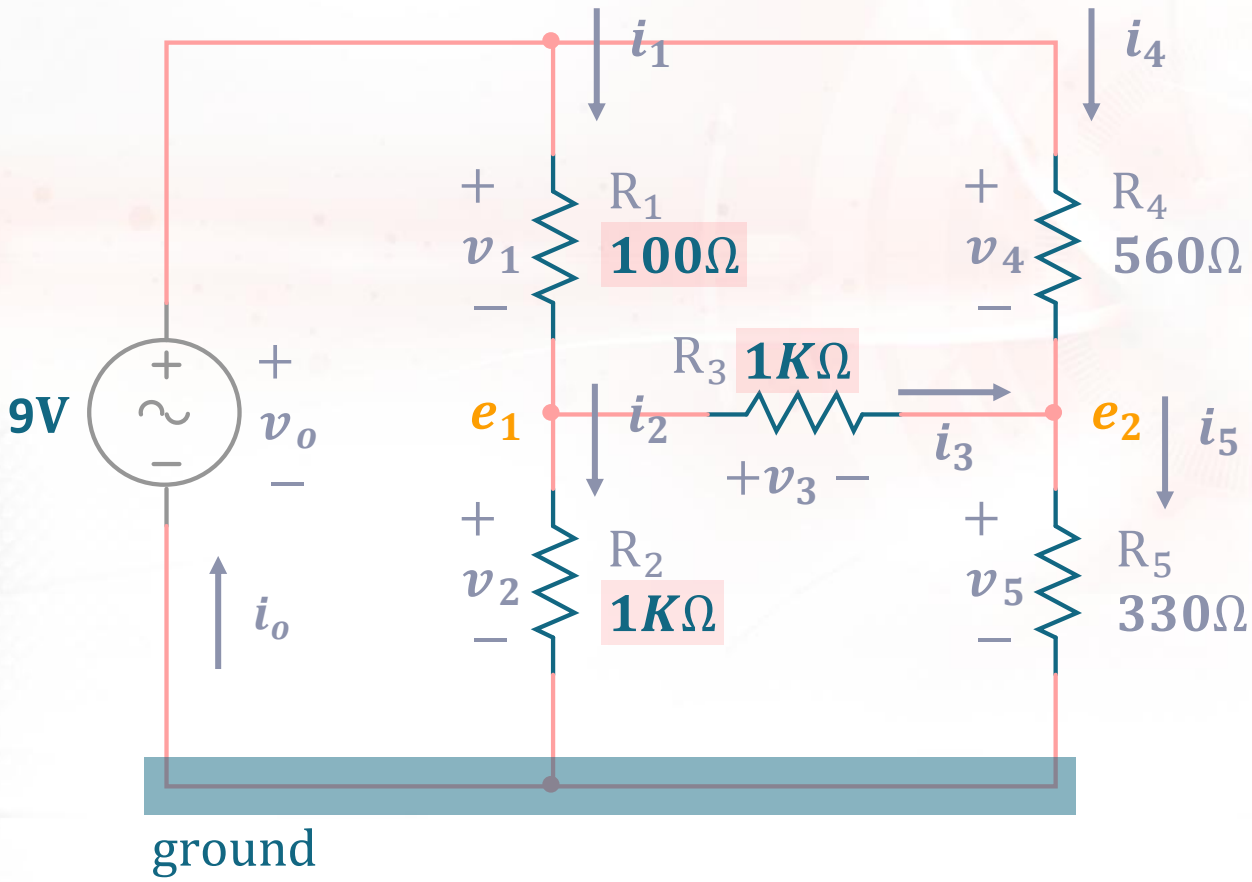
$$-\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = 0$$

$$-\frac{v_o - e_1}{R_1} + \frac{e_1 - 0}{R_2} + \frac{e_1 - e_2}{R_3} = 0$$

$$-\frac{v_o}{R_1} + \frac{e_1}{R_1} + \frac{e_1}{R_2} + \frac{e_1}{R_3} - \frac{e_2}{R_3} = 0$$

$$-\frac{v_o}{R_1} + e_1\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}\right) - \frac{e_2}{R_3} = 0$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



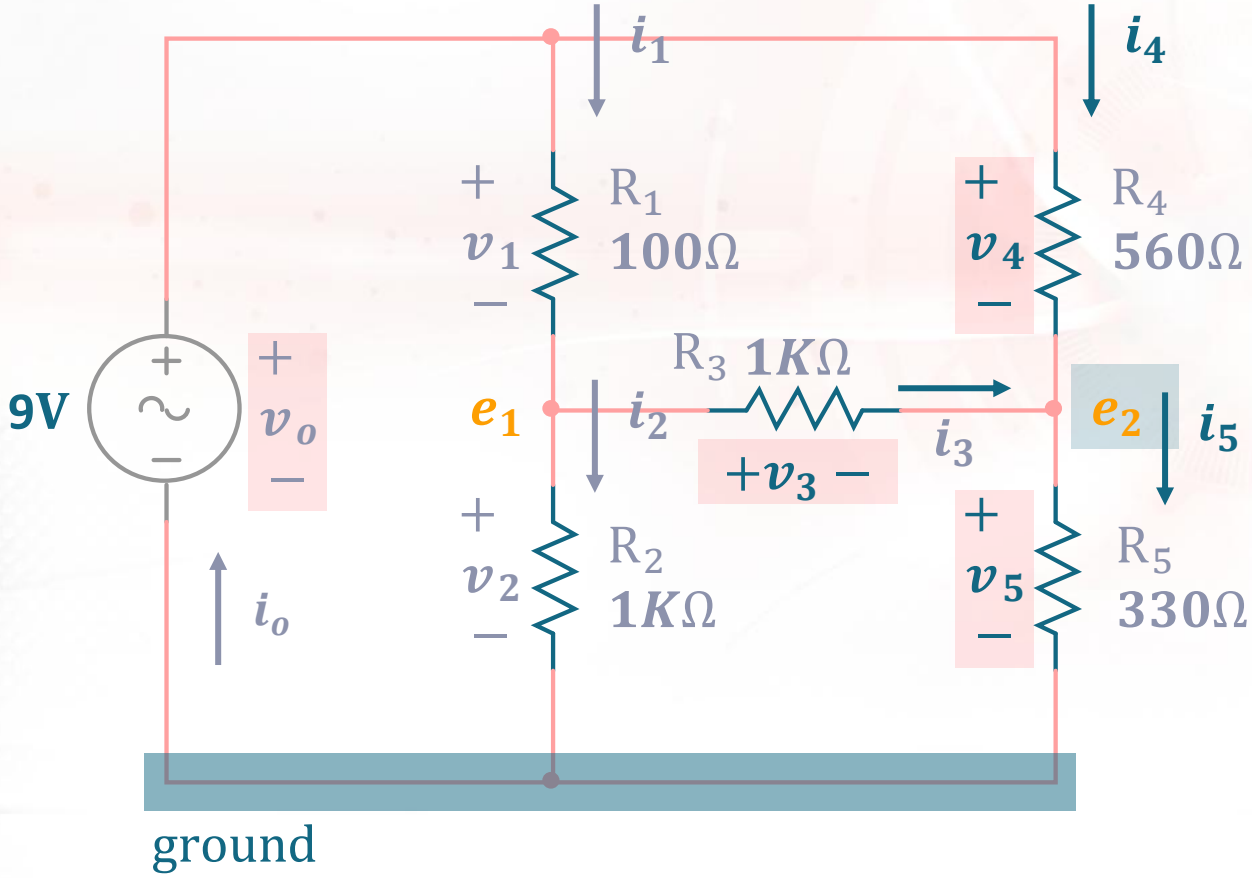
$$-\frac{v_o}{R_1} + e_1\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right) - \frac{e_2}{R_3} = 0$$

$$e_1\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right) - \frac{e_2}{R_3} = \frac{v_o}{R_1}$$

$$e_1\left(\frac{1}{100}+\frac{1}{1K}+\frac{1}{1K}\right) - \frac{e_2}{1K} = \frac{9}{100}$$

1 $$\frac{3}{250}e_1 - \frac{1}{1K}e_2 = \frac{9}{100}$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



KCL @ e_2

$$-i_3 - i_4 + i_5 = 0$$

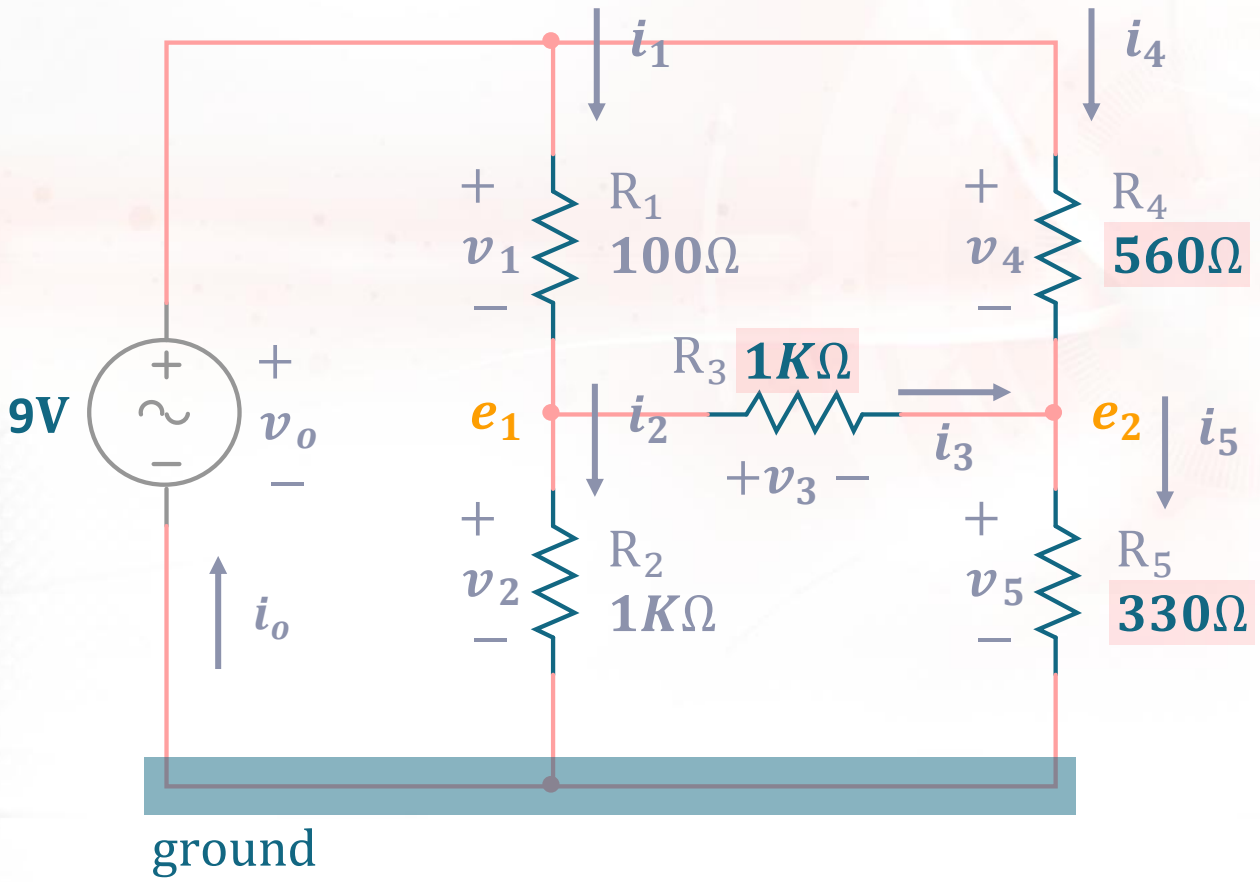
$$-\frac{v_3}{R_3} - \frac{v_4}{R_4} + \frac{v_5}{R_5} = 0$$

$$-\frac{e_1 - e_2}{R_3} - \frac{v_o - e_2}{R_4} + \frac{e_2 - 0}{R_5} = 0$$

$$-\frac{e_1}{R_3} + \frac{e_2}{R_3} - \frac{v_o}{R_4} + \frac{e_2}{R_4} + \frac{e_2}{R_5} = 0$$

$$-\frac{v_o}{R_4} + e_2\left(\frac{1}{R_3} + \frac{1}{R_4} + \frac{1}{R_5}\right) - \frac{e_1}{R_3} = 0$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



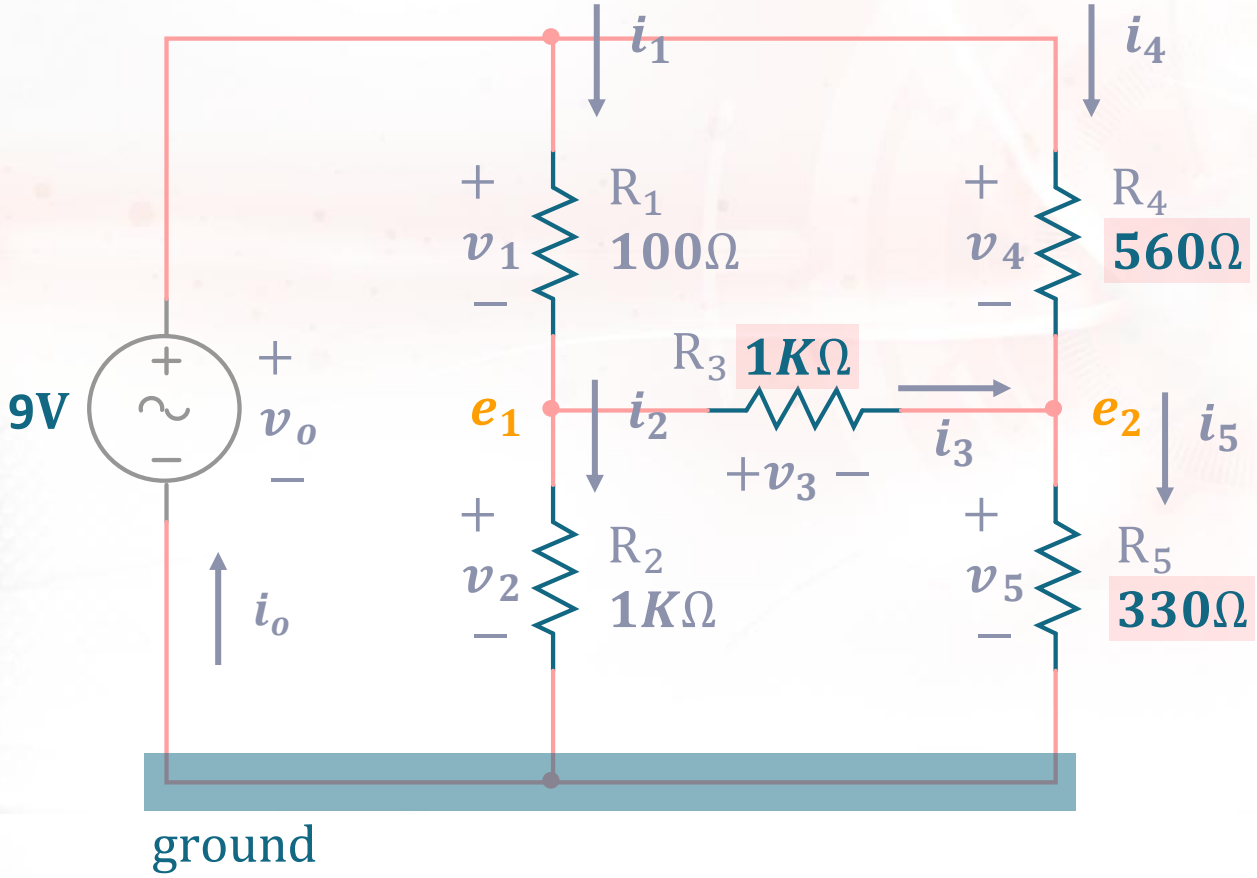
$$-\frac{v_o}{R_4} + e_2\left(\frac{1}{R_3}+\frac{1}{R_4}+\frac{1}{R_5}\right) - \frac{e_1}{R_3} = 0$$

$$-\frac{e_1}{R_3} + e_2\left(\frac{1}{R_3}+\frac{1}{R_4}+\frac{1}{R_5}\right) = \frac{v_o}{R_4}$$

$$-\frac{e_1}{1K} + e_2\left(\frac{1}{1K}+\frac{1}{560}+\frac{1}{330}\right) = \frac{9}{560}$$

2 $$-\frac{1}{1K}e_1 + \frac{2687}{462K}e_2 = \frac{9}{560}$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



Linear Equations

$$\frac{3}{250}e_1 - \frac{1}{1K}e_2 = \frac{9}{100}$$

$$-\frac{1}{1K}e_1 + \frac{2687}{462K}e_2 = \frac{9}{560}$$

Augmented Matrix

$$\begin{array}{cc} & \begin{array}{ccc} e_1 & e_2 & \end{array} \\ & \begin{bmatrix} \frac{3}{250} & -\frac{1}{1K} & \frac{9}{100} \\ -\frac{1}{1K} & \frac{2687}{462K} & \frac{9}{560} \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \end{array} \end{array}$$

Linear Equations

$$\frac{3}{250}e_1 - \frac{1}{1K}e_2 = \frac{9}{100}$$

$$-\frac{1}{1K}e_1 + \frac{2687}{462K}e_2 = \frac{9}{560}$$

Augmented Matrix

$$\begin{array}{cc} \begin{array}{ccc} e_1 & e_2 & \end{array} & \\ \begin{bmatrix} \frac{3}{250} & -\frac{1}{1K} & \frac{9}{100} \\ -\frac{1}{1K} & \frac{2687}{462K} & \frac{9}{560} \end{bmatrix} & \begin{array}{c} r_1 \\ r_2 \end{array} \end{array}$$

$$\begin{array}{cc} \begin{array}{ccc} e_1 & e_2 & \end{array} & \\ \begin{bmatrix} \frac{3}{250} & -\frac{1}{1K} & \frac{9}{100} \\ 0 & \frac{5297}{77M} & \frac{99}{350K} \end{bmatrix} & \begin{array}{c} r_1 \\ r_2 \end{array} \end{array}$$

$$r_2 \leftarrow \frac{1}{1K} r_1 + \frac{3}{250} r_2$$

From equation 2

$$\frac{5297}{77M}e_2 = \frac{99}{350K}$$

$$e_2 = \frac{99(77M)}{350K(5297)}$$

$$e_2 = 4.11\ V$$

$$\begin{array}{cc} & \begin{array}{ccc} e_1 & e_2 & \end{array} \\ & \begin{bmatrix} \frac{3}{250} & -\frac{1}{1K} & \frac{9}{100} \\ 0 & \frac{5297}{77M} & \frac{99}{350K} \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \end{array} \end{array}$$

From equation 1

$$\frac{3}{250}e_1 - \frac{1}{1K}e_2 = \frac{9}{100}$$

$$\frac{3}{250}e_1 - \frac{1}{1K}(4.11) = \frac{9}{100}$$

$$\frac{3}{250}e_1 = \frac{9}{100} + \frac{4.11}{1K}$$

$$\frac{3}{250}e_1 = \frac{9411}{100K}$$

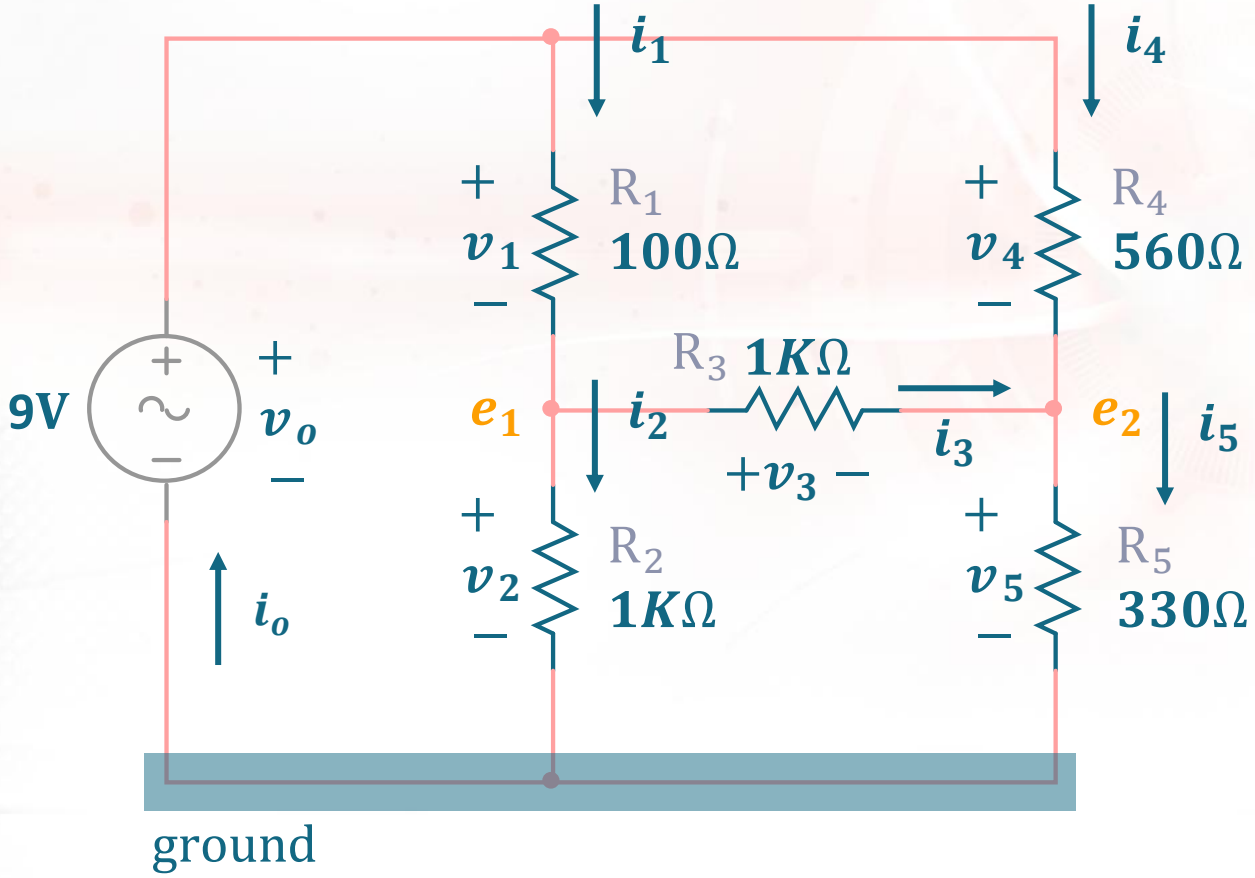
$$e_1 = \frac{9411(250)}{100K(3)}$$

$$e_1 = 7.84\ V$$

$$e_2 = 4.11\ V$$

$$\begin{array}{cc} & \begin{array}{ccc} e_1 & e_2 & \\ \end{array} \\ & \begin{bmatrix} \frac{3}{250} & -\frac{1}{1K} & \frac{9}{100} \\ 0 & \frac{5297}{77M} & \frac{99}{350K} \end{bmatrix} \begin{array}{c} r_1 \\ r_2 \end{array} \end{array}$$

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$e_1 = 7.84\ V$ $\qquad$ $e_2 = 4.11\ V$

$v_1 = v_o - e_1$

$v_1 = 9 - 7.84$

$\boxed{v_1 = 1.16\ V}$

ans

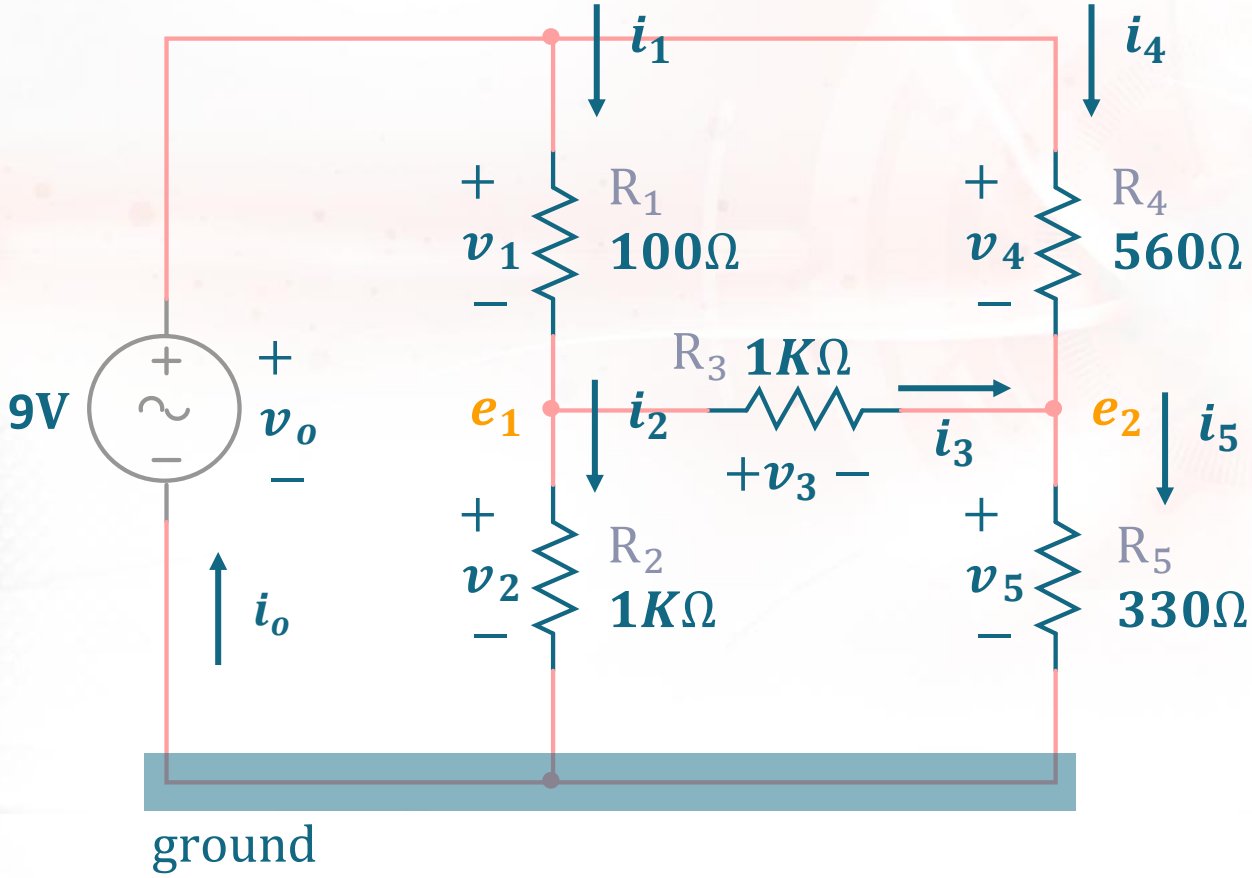
$i_1 = \dfrac{v_1}{R_1}$ $\qquad$ $P_1 = v_1 i_1$

$i_1 = \dfrac{1.16}{100}$ $\qquad$ $P_1 = (1.16)(11.6 \times 10^{-3})$

$\boxed{i_1 = 11.6\ mA}$ $\qquad$ $\boxed{P_1 = 13.46\ mW}$

ans $\qquad$ *ans*

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$e_1 = 7.84\ V$

$e_2 = 4.11\ V$

$v_2 = e_1$

$v_2 = 7.84\ V$

ans

$i_2 = \frac{v_2}{R_2}$

$P_2 = v_2 i_2$

$i_2 = \frac{7.84}{1K}$

$P_2 = (7.84)(7.84 \times 10^{-3})$

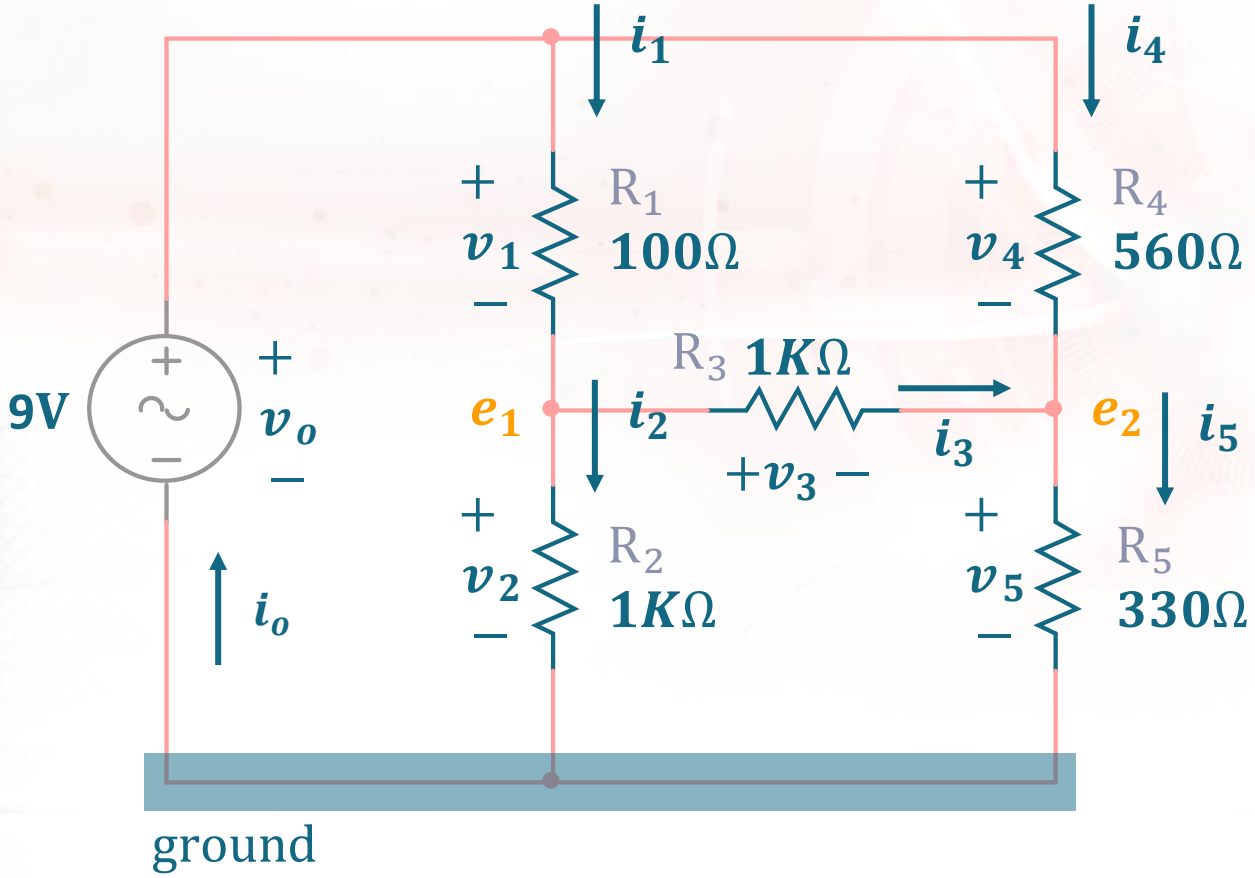
$i_2 = 7.84\ mA$

ans

$P_2 = 61.47\ mW$

ans

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$e_1 = 7.84\ V$ $\qquad$ $e_2 = 4.11\ V$

$$v_3 = e_1 - e_2$$

$$v_3 = 7.84 - 4.11$$

$$\boxed{v_3 = 3.73\ V}$$

ans

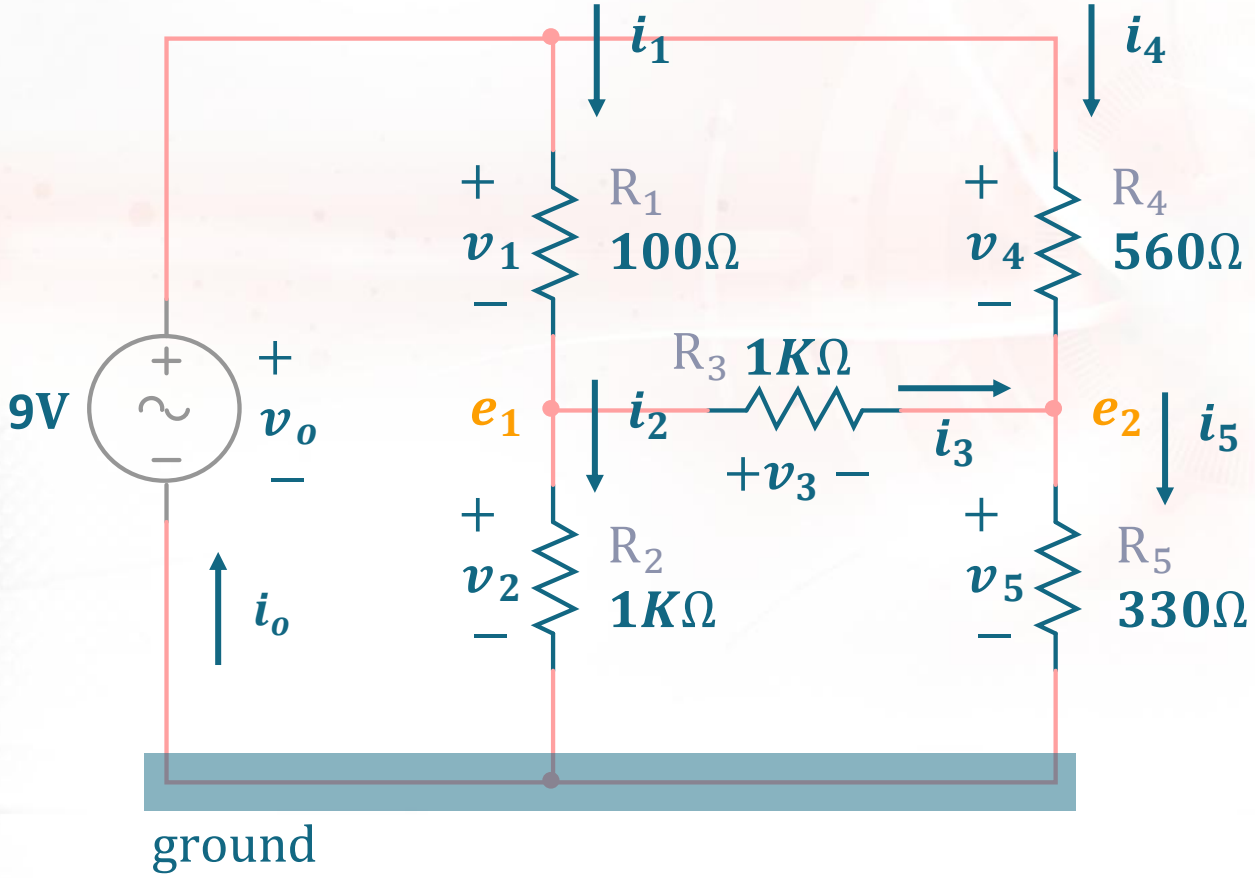
$$i_3 = \frac{v_3}{R_3} \qquad P_3 = v_3 i_3$$

$$i_3 = \frac{3.73}{1K} \qquad P_3 = (3.73)(3.73 \times 10^{-3})$$

$$\boxed{i_3 = 3.73\ mA} \qquad \boxed{P_3 = 13.91\ mW}$$

ans $\qquad$ *ans*

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$e_1 = 7.84\ V$ $\qquad$ $e_2 = 4.11\ V$

$v_4 = v_o - e_2$

$v_4 = 9 - 4.11$

$\boxed{v_4 = 4.89\ V}$

ans

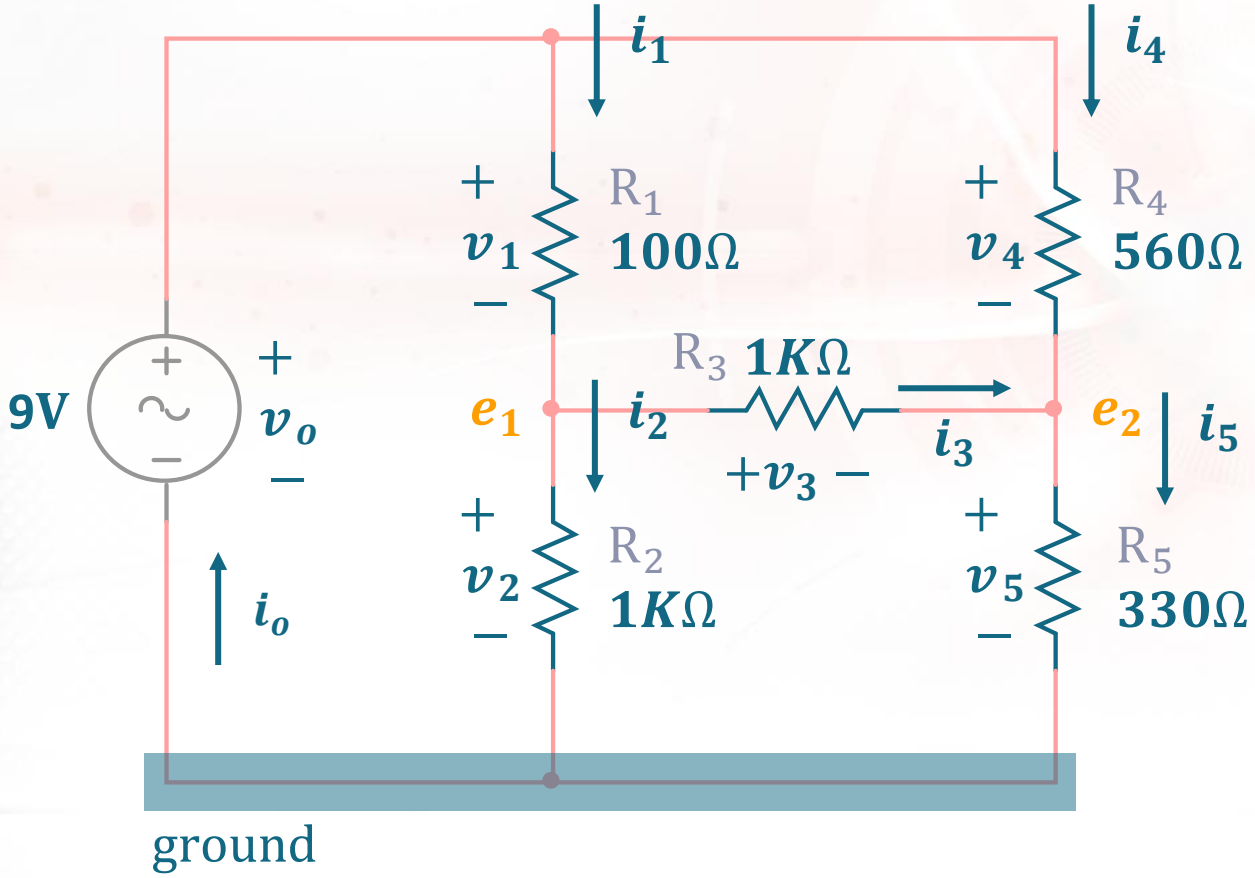
$i_4 = \dfrac{v_4}{R_4}$ $\qquad$ $P_4 = v_4 i_4$

$i_4 = \dfrac{4.89}{560}$ $\qquad$ $P_4 = (4.89)(8.73 \times 10^{-3})$

$\boxed{i_4 = 8.73\ mA}$ $\qquad$ $\boxed{P_4 = 42.69\ mW}$

ans $\qquad$ *ans*

Determine the value of **current**, **voltage**, and **power** of every element in the given circuit using **node analysis** method.



$e_1 = 7.84\ V$

$e_2 = 4.11\ V$

$v_5 = e_2$

$v_5 = 4.11\ V$

ans

$i_5 = \frac{v_5}{R_5}$

$P_5 = v_5 i_5$

$i_5 = \frac{4.11}{330}$

$P_5 = (4.11)(12.45 \times 10^{-3})$

$i_5 = 12.45\ mA$

ans

$P_2 = 51.17\ mW$

ans

LABORATORY